

Supplementary Information for EFF-Dock: Fragment-Based SE(3)-Equivariant Flow Matching for Protein–Ligand Docking

Jaemin Sim¹ and Juyong Lee^{1,2,3,4,*}

¹Department of Molecular Medicine and Biopharmaceutical Sciences, Graduate School of Convergence Science and Technology, Seoul National University, Seoul 08826, Republic of Korea

²College of Pharmacy, Seoul National University, Seoul 08826, Republic of Korea

³Research Institute of Pharmaceutical Science, College of Pharmacy, Seoul National University, Seoul 08826, Republic of Korea

⁴Arontier Co., Seoul 06735, Republic of Korea

*Corresponding author: nicole23@snu.ac.kr

This document provides supplementary methods, implementation details, evaluation protocols, and additional analyses for the accompanying article.

S1 Task Definition and Input Preparation

S1.1 Notation and prediction task

Each prediction uses a prepared receptor structure, one ligand chemical state and conformer, and a supplied pocket definition. The receptor remains fixed, and the model predicts fragment translations and proper rotations. Table S1 summarizes the notation used throughout the article and Supplementary Information.

Table S1: Notation for ligand pose generation and sampling.

Symbol	Definition
$\mathcal{P}$	Fixed receptor atom/residue graph obtained from the prepared receptor and specified crop.
$\mathcal{G}_L$	Hydrogen-stripped ligand graph after retaining the largest connected component; indexed atom order, chemistry and stereochemistry define its chemical identity.
$\mathcal{C}_L$	Supplied or generated input conformer from which fragment-local coordinates are prepared.
N, K	Numbers of ligand heavy atoms and rigid fragments, respectively; i and k index atoms and fragments.
$\mathcal{F} = \{1, \dots, K\}$	Deterministic rigid-fragment partition; $g(i)$ maps ligand atom i to a fragment.
$\mathbf{y}_i$	Fixed coordinate of atom i in its fragment-local conformer frame.
$\mathbf{z}_t$	Fragment pose state $\{(\mathbf{T}_{k,t}, \mathbf{R}_{k,t})\}_{k=1}^K$ in the pocket-centered world frame, with $\mathbf{T}_{k,t} \in \mathbb{R}^3$ and $\mathbf{R}_{k,t} \in \text{SO}(3)$.
$\mathbf{x}_{i,t}$	Reconstructed atom position $\mathbf{R}_{g(i),t}\mathbf{y}_i + \mathbf{T}_{g(i),t}$.
$\mathbf{b}$	Supplied pocket center. For prospective use, it must be specified independently of the unknown ligand pose.
M, S, σ	Number of sampled candidates, number of integration intervals, and translation-prior scale, respectively.

Units, operators and reused indices. Bold symbols denote vectors or matrices; calligraphic symbols denote sets, graphs or state spaces. $\mathbb{R}^3$, $\text{SO}(3)$ and $\text{SE}(3)$ denote Euclidean three-space,

proper three-dimensional rotations, and their translation-rotation group. $\mathbf{I}_3$ is the identity matrix, $\top$ is transpose, $^{-1}$ is inverse, and $^{+}$ is the specified truncated pseudoinverse. $|\mathcal{A}|$ is set cardinality, $\|\cdot\|_2$ the Euclidean norm, $\times$ the vector cross product, $\odot$ element-wise multiplication, and $\oplus$ feature concatenation in the architecture diagram. Stars indicate targets or an explicitly defined selected index; hats indicate predictions, except that $\hat{\mathbf{h}}$ denotes normalized hidden features in Fig. 2.

Coordinates, distances, displacement and RMSD use Angstrom; angles use radians unless degrees are displayed. Flow time $t \in [0, 1]$ is dimensionless transport time, not elapsed simulation time. $m = 1, \dots, M$ indexes candidates, $s = 0, \dots, S$ integration states, and b a minibatch complex; bold $\mathbf{b}$ instead denotes the pocket center. The N in the pose representation counts heavy atoms; N in candidate-prefix plots and workflow labels instead denotes the available candidate count. n in benchmark labels counts complexes or explicitly labelled strata. A top- k cutoff is a ranking count, not a fragment index. All logarithms are natural and act on numerical values in the stated units: $\log \sigma$ abbreviates $\log[\sigma/(1 \text{ \AA})]$, and $\log(1 + r)$ abbreviates $\log[1 + r/(1 \text{ \AA})]$ for an RMSD r . Logits are real pre-sigmoid scores; $\text{sigmoid}(a) = 1/(1 + e^{-a})$ and $\text{softplus}(a) = \log(1 + e^a)$ for scalar argument a . Hidden features have no assigned physical units. The rotation Exp map converts an axis-angle vector to a proper rotation, and Log returns its principal axis-angle vector. $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes a Gaussian with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$; $\mathbb{I}[A]$ is one when condition A holds and zero otherwise. Diagram $\otimes$ denotes an equivariant tensor product.

Numerical values denote measured outcomes, benchmark criteria, fixed settings or numerical safeguards. Fixed settings reproduce the evaluated implementation; their optimality requires an explicit selection criterion or controlled experiment.

Input standardization and coordinate frame. Geometry-training ligands are read from crystal-complex coordinates with RDKit, and the sanitization outcome is recorded [1]. Stereochemistry is assigned from coordinates before hydrogen removal, and the largest connected component defines the heavy-atom graph. For SMILES input, one ETKDGv3 conformer is generated with the protocol-specific seed [2] and optimized for up to 200 MMFF iterations [3]; coordinates supplied in SDF or MOL2 format are retained. Formal charge, stereochemistry, and protonation state are preserved without tautomer or protonation-state enumeration.

The receptor parser removes hydrogens, waters, nucleic acids, and non-primary alternate locations; normalizes supported residue variants; reconstructs standard intra-residue, peptide, and disulfide connectivity; and retains recognized monatomic metals as typed nodes. For all reported refinements and guidance controls, unresolved receptor chemistry contributes bounded repulsion without attractive interaction typing. The neural graph uses a normalized receptor crop, whereas the optional guidance calculation uses a larger PDB-derived shell. Both receptor and ligand coordinates are translated by $-\mathbf{b}$ so that the pocket center is the origin.

For geometry training, fragments are constructed from the crystal ligand pose. Fragment-local coordinates are centered at the corresponding geometric centroids, making the unaugmented target rotation the identity. During augmentation, a common proper rotation is applied to the local coordinates and its inverse becomes the target rotation. Confidence training used a proper-rotation Kabsch fit to recover fragment poses [4]. This historical recovery excludes reflections and leaves singleton rotations at identity; it does not resolve the unobservable twist of linear fragments. Evaluation-specific recovery checks are described in Sec. S1.3. Rotational masking is applied separately in the Newton-Euler readout and angular-velocity loss. World-frame equivariance tests hold these prepared local templates fixed and jointly transform the receptor, candidate coordinates and fragment states. They do not test invariance to reorienting the supplied conformer before fragmentation. Fragment centering retains the conformer’s body-axis convention rather than imposing a canonical orientation.

S1.2 Coordinate convention and evaluated weights

Fragment reconstruction uses the active local-to-world convention $\mathbf{x} = \mathbf{R}\mathbf{y} + \mathbf{T}$. For a fragment-local learned vector $\mathbf{w}$, the world-frame injection is

$$F(\mathbf{R}, \mathbf{w}) = \mathbf{R}\mathbf{w}, \quad F(\mathbf{Q}\mathbf{R}, \mathbf{w}) = \mathbf{Q}F(\mathbf{R}, \mathbf{w}). \quad (\text{S1})$$

Source-local edge coordinates $\mathbf{R}^\top(\mathbf{x}_j - \mathbf{x}_i)$ map a world displacement into a local frame.

Evaluation uses fixed docking and confidence weights with local-to-world feature injection and coordinate-to-frame recovery that differ from training. These conventions also determine the terminal docking features supplied to the confidence network.

Finite network verification used five complexes, three flow times per complex, and four joint rotations per time. All 60 tested translation and 60 angular outputs satisfied the tolerance $\|\Delta\|_2 \leq 10^{-4} + 10^{-4}\|\text{expected}\|_2$; the maximum relative errors were 9.2×10^{-6} and 1.8×10^{-5} , respectively. These core-network tests support covariance under joint world rotations but do not prove exact equivariance of the entire data-preparation, sampling, refinement, and selection pipeline. In the tolerance expression, Δ is measured output minus the jointly transformed expected output. The two 10^{-4} coefficients are absolute and relative numerical acceptance tolerances, rather than docking-accuracy thresholds. Singular-value floors, anchor-angle tolerances and inertia truncation likewise identify numerically usable geometry; they are not learned parameters.

S1.3 Numerical checks for coordinate-only frame recovery

Coordinate-based recovery leaves the axial twist of linear fragments undetermined. Although this twist does not change fragment coordinates, it affects terminal docking features and confidence scores; a consistent recovery convention therefore does not establish twist invariance.

Primary measurements use a context-defined recovery convention that leaves the proper Kabsch fit unchanged for templates of numerical rank at least two. Rank is computed from centered-template singular values using a relative tolerance of 10^{-6} and an absolute floor of 10^{-12} . For rank-one fragments, the observed axis is retained, and an atom-indexed, nonparallel template-pose context pair fixes the unobservable twist. When paired context is unavailable, a fixed template-local basis supplies the body direction and posed ligand atoms, followed by receptor atoms, supply its world counterpart. Singleton frames use two nonparallel scene vectors. No global world-axis fallback is used; collapsed or fully collinear inputs are rejected. Anchor indexing is a fixed input convention, not an atom-permutation-invariance claim.

A context vector is accepted when its perpendicular component exceeds 10^{-5} times the larger of its length and $1 \text{ \AA}$. The first accepted atom in input order is used rather than the best-conditioned direction. Because this anchor can belong to another fragment, the recovered twist can change when that fragment moves, even if the observed linear fragment does not. This is a frame convention rather than a rigid-body observable. Nearly collinear templates above the rank tolerance retain Kabsch recovery, for which these tests do not establish uniform numerical stability.

The transformation tests kept the prepared body template, atom order, graph definitions and model parameters fixed. They jointly rotated receptor and candidate coordinates, then reconstructed candidate frames afresh. Five prescribed rotations were applied to the first saved raw pose of each of Astex 1S19, 1W2G, 1YGC and the rank-at-least-two control 1GM8; a separate joint translation was also tested. All contact identities were retained. Across the 20 rotation tests, the maximum predicted-RMSD change was $2.4 \times 10^{-6} \text{ \AA}$ and the maximum terminal-feature covariance residual, expressed as a relative Euclidean norm, was 1.6×10^{-6} . The four translation tests changed predicted RMSD by at most $7.2 \times 10^{-7} \text{ \AA}$. These finite tests support covariance under a joint world transformation conditional on the prepared body convention. They do not prove invariance to input-conformer reorientation, atom reordering, scene changes or the complete docking pipeline.

Separately sweeping the unobservable twist through 0° , 90° , 180° and 270° in the three rank-one examples changed predicted RMSD over ranges of 8.5×10^{-4} to $1.5 \times 10^{-3} \text{ \AA}$. Fixing this twist consistently therefore does not make the frozen score invariant to it.

Table S2 compares CPU and GPU frame recovery on two fixed 100-pose raw and refined banks. Network arithmetic, coordinates, weights and other inputs were held on the same A5000 GPU. The context-defined convention reduced the measured numerical discrepancy. This tests frame-recovery consistency rather than docking accuracy.

S1.4 Pocket definition and receptor cropping

For PLINDER complexes, the stored center is the centroid of receptor residue nodes within $8 \text{ \AA}$ of the deposited ligand; the ligand centroid is used when no residue node satisfies this criterion [5]. Residue-aware cropping retains an entire residue when any of its atoms lies within the cutoff. Table S3 lists the crop used for each experimental setting.

Table S2: Recovery-backend sensitivity with model arithmetic and coordinates fixed. Each row covers all 200 candidates from one raw and one refined bank. Values are maximum absolute predicted-RMSD differences in Angstrom; no crystal labels or docking outcomes determine these tests.

Bank	Kabsch recovery	Context-defined recovery	Same-GPU repeat
OpenBind a71ev2a-x0152a	9.3×10^{-3}	5.2×10^{-6}	5.0×10^{-6}
FoldBench 9ENH, FAD	1.5×10^{-3}	1.4×10^{-6}	1.2×10^{-6}

Table S3: Task-specific pocket-center, crop, and empty-crop provenance.

Context	Center available to the pipeline	First crop	Empty-crop policy
Geometry training	Stored crystal-derived center plus $\epsilon \sim \mathcal{N}(\mathbf{0}, 2^2 \mathbf{I}_3)$ Å	$r \sim \text{Uniform}(6, 12)$ Å about the jittered center	Retry about theunjittered stored center at 13 Å, then 40 Å, then retain the nearest 32 residues
Validation	Stored crystal-derived center; no jitter	Fixed 8 Å	Same ordered 13 Å $\rightarrow$ 40 Å $\rightarrow$ nearest-32 fallback
External reference-pocket redocking	Reference-derived center computed from residue nodes within 8 Å of the reference ligand, with ligand-centroid fallback; the center is fixed before the ligand is removed from the input	Fixed 10 Å; no jitter	An empty crop is recorded as a complex-level failure
Prospective known-pocket docking	User- or predictor-supplied center or residues, independent of the unknown pose	Specified before evaluation	Specified and reported for the application

The external experiments are reference-pocket holo redocking rather than blind docking. Reference-ligand coordinates define the pocket once and are then used only for evaluation.

S2 Fragmentation and Flow Sampling

S2.1 Exact rigid-fragment decomposition

Input processing accepts full sanitization or a recorded partial-sanitization fallback (`sanitize_ok`). Fragmentation is an indexed graph construction rather than RDKit’s generic rotatable-bond SMARTS. For an H-stripped ligand graph $G = (V, E)$, let $\chi_{ij} = 1$ when (i, j) is a single, non-ring bond, both atoms have heavy-atom degree greater than one, and the bond is neither a planar C(=O)–N/O nor S(=O)₂–N linkage; biaryl aromatic–aromatic single bonds remain eligible. The lexicographically ordered candidate set is

$$\mathcal{C}_0 = \text{sort}_{\text{lex}}\{(\min(i, j), \max(i, j)) \in E : \chi_{ij} = 1\}. \quad (\text{S2})$$

Starting from $r_i^{(0)} = \deg_G(i)$, candidates are swept once in that order. Candidate $e_q = (i, j)$ is retained iff $r_i^{(q-1)} > 1$ and $r_j^{(q-1)} > 1$; then

$$r_a^{(q)} = r_a^{(q-1)} - \mathbb{I}[a \in \{i, j\}] \mathbb{I}[e_q \text{ is retained}]. \quad (\text{S3})$$

Here q is the candidate-bond sweep index, $a \in V$ is an arbitrary atom index, and $\mathbb{I}[\cdot]$ is the indicator function. The vertex and edge sets V, E are heavy atoms and covalent bonds; $\deg_G(i)$ is heavy-atom degree. Eligibility χ_{ij} is zero when the stated rule fails, sort_{lex} orders atom-index pairs lexicographically, and $r_i^{(q)}$ is the remaining degree after the first q candidate cuts. Removing the retained cuts gives the preliminary connected components. Singleton components are repeatedly merged into the largest adjacent component (with RDKit neighbor order resolving a tie), until no singleton remains or a sweep makes no change. Cuts whose endpoints subsequently share a component are discarded. For the resulting atom set $\mathcal{A}_k$,

$$\mathbf{T}_k = \frac{1}{|\mathcal{A}_k|} \sum_{i \in \mathcal{A}_k} \mathbf{x}_i, \quad \mathbf{y}_i = \mathbf{x}_i - \mathbf{T}_{g(i)}. \quad (\text{S4})$$

Non-finite centers or local coordinates yield a decomposition failure. The retained cuts define the bidirectional cut and fragment-adjacency relations; cross-cut one-hop atom pairs define the triangulation relations and retain their reference distances.

The algorithm is deterministic for a fixed atom-indexed molecule, although the lexicographic sweep can change under atom reindexing. The training loader accepts at most 30 fragments; this is a dataset restriction rather than a limit of the decomposition.

Fragment-local conformers. Training uses local coordinates from bound ligands. At inference, local coordinates come from supplied or independently generated conformers. Rigid-fragment motion cannot correct internal geometric differences within a fragment.

Flow coupling and training-time sampling. After pocket centering, the conditional prior factorizes across fragments as

$$\mathbf{T}_{k,0} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_3), \quad \mathbf{R}_{k,0} \sim \text{Uniform}(\text{SO}(3)), \quad (\text{S5})$$

with singleton rotations fixed to identity; multi-atom rotations are sampled from the normalized Haar distribution using a uniform-quaternion construction [6]. Geometry training samples

$$\sigma \in \{0.5, 1, 2, 3, 4\} \text{ \AA}, \quad p_\sigma = \{0.10, 0.25, 0.30, 0.25, 0.10\}, \quad (\text{S6})$$

and embeds $\log \sigma$ together with flow time. The entries of p_σ correspond in order to the five listed scales and sum to one; they specify the training frequency of each spatial prior. For paired prior and target fragment poses, the left-action coupling is

$$\begin{aligned} \mathbf{T}_{k,t} &= (1-t)\mathbf{T}_{k,0} + t\mathbf{T}_{k,1}, & \mathbf{v}_k^* &= \mathbf{T}_{k,1} - \mathbf{T}_{k,0}, \\ \boldsymbol{\omega}_k^* &= \text{Log}(\mathbf{R}_{k,1}\mathbf{R}_{k,0}^{-1}), & \mathbf{R}_{k,t} &= \text{Exp}(t\boldsymbol{\omega}_k^*)\mathbf{R}_{k,0}. \end{aligned} \quad (\text{S7})$$

The implementation represents rotations by scalar-first unit quaternions, flips the target endpoint only when its inner product with the prior endpoint is negative, and retains the stored endpoint sign when that inner product is exactly zero. Consequently, the relative quaternion selects a deterministic numerical axis branch for the measure-zero exact- π case before conversion to $\boldsymbol{\omega}_k^*$. A common random proper rotation augments the ligand representation. The base logit-normal-mixture time distribution is

$$p_{\text{base}}(t) = 0.02\mathcal{U}(0, 1) + 0.98 \text{Law}[\text{sigmoid}(0.8 + 1.7Z)], \quad Z \sim \mathcal{N}(0, 1). \quad (\text{S8})$$

The released docking checkpoint uses

$$p_{\text{fit}}(t) = 0.80p_{\text{base}}(t) + 0.10\mathcal{U}(0, 0.3) + 0.10\delta_0. \quad (\text{S9})$$

The δ_0 component replays the exact prior endpoint at $t = 0$. Here $\mathcal{U}(a, b)$ is uniform on $[a, b]$, $\text{Law}[X]$ denotes the distribution of X , and δ_0 is the unit point mass at zero. The 0.8 and 1.7 in the base distribution are the location and scale of the Gaussian before the sigmoid. The 0.02 uniform component preserves sampling across the full interval; the continuation adds early-time draws and exact-prior replay. These parameters are retained from the training configuration. At inference, the sampler uses

$$t_s = 1 - \left(1 - \frac{s}{S}\right)^3, \quad s = 0, \dots, S, \quad (\text{S10})$$

with $\Delta t_s = t_{s+1} - t_s$, the explicit Euler translation update $\mathbf{T}_{k,t+\Delta t} = \mathbf{T}_{k,t} + \Delta t \hat{\mathbf{v}}_{k,t}$, and the left-multiplicative rotation update below. The power 3 makes intervals shorter near the target endpoint; S intervals use $S + 1$ grid points including both endpoints. The grid power and primary ten-interval budget are fixed protocol choices.

$$\mathbf{R}_{k,t+\Delta t} = \text{Exp}(\Delta t \hat{\boldsymbol{\omega}}_{k,t}) \mathbf{R}_{k,t}. \quad (\text{S11})$$

S2.2 End-to-end candidate generation and selection

The fragment construction, receptor crop, and static heterogeneous graph define the initial state for each receptor–ligand pair. The M candidates are sequential draws from the prior in Eq. (S5); candidate indices denote draw order rather than independently seeded streams. At interval s , protein–ligand contacts within 5 Å are rebuilt and the flow network predicts fragment fields $(\hat{\mathbf{v}}_s^{(m)}, \hat{\boldsymbol{\omega}}_s^{(m)})$. With $\Phi_{\Delta t_s}$ denoting the SE(3) integration step, sampling is

$$\mathbf{z}_{s+1}^{(m)} = \Phi_{\Delta t_s}(\mathbf{z}_s^{(m)}; \hat{\mathbf{v}}_s^{(m)}, \hat{\boldsymbol{\omega}}_s^{(m)}), \quad s = 0, \dots, S - 1, \quad (\text{S12})$$

Raw endpoints are saved before refinement. For a confidence model c_ϕ , the selected pose is

$$\mathcal{E} = \begin{cases} \{m : q_m = 1\}, & \text{if any } q_m = 1, \\ \{1, \dots, M\}, & \text{otherwise,} \end{cases} \quad m^* = \arg \min_{m \in \mathcal{E}} c_\phi(\mathbf{z}_S^{(m)}, \mathcal{P}, \mathcal{G}_L, \mathcal{C}_L, \sigma). \quad (\text{S13})$$

Here $q_m = 1$ means that candidate m matches all assigned input tetrahedral stereocenters, and $q_m = 0$ otherwise; $\mathcal{E}$ is the effective eligible set, including the all-ineligible fallback. Ties follow saved candidate order. The unfiltered argmin is reported separately as a diagnostic condition. The function c_ϕ is the nonnegative predicted RMSD from Eq. (S20), parameterized by learned weights ϕ . Unsupported inputs and sampling exceptions are recorded as complex-level failures.

S3 Equivariant Architecture and Objectives

This section records the architectural choices needed to interpret the model and reproduce its outputs; low-level tensor-product paths and parameter tensors remain defined by the released configuration and source code. The use of irreducible features and tensor-product message passing is based on the Equiformer representation framework [7]. Equivariant linear, tensor-product and spherical-harmonic operations in the evaluated implementation use cuEquivariance 0.9.0 [8], with an e3nn-compatible multiplicity–irrep feature layout [9]. All distances passed to neural networks are represented numerically in Angstrom. Primary measurements use the local-to-world fragment feature injection in Eq. (S1); local-frame edge displacements use the inverse world-to-local operation.

S3.1 Equivariant trunk and graph

The docking network applies six shared message-passing layers to ligand atoms, ligand fragments, protein atoms, and residue virtual nodes. Each hidden state contains 384 even scalar channels, 32 polar-vector channels, 32 axial-vector channels, and 16 even- and 16 odd-parity rank-2 channels. Flow time and the logarithm of the translation-prior scale are encoded separately and summed to form a 128-dimensional conditioning vector $\boldsymbol{\eta}_b$ for complex b . For edge $j \rightarrow i$, a compact layer description is

$$\mathbf{h}_i^{(\ell+1)} = \text{AdaLN}_{\boldsymbol{\eta}_{b(i)}} \left[\mathbf{h}_i^{(\ell)} + \text{Block}^{(\ell)} \left(\text{GateMean}_{j \in \mathcal{N}(i)} \text{Message}^{(\ell)} \left(\text{Norm}(\mathbf{h}_j^{(\ell)}), \mathbf{Y}_{0:2}(\hat{\mathbf{r}}_{ji}), \mathbf{e}_{ji}^{(\ell)} \right) \right) \right]. \quad (\text{S14})$$

Here $\mathbf{Y}_{0:2}$ denotes real spherical harmonics through order two and $\mathbf{e}_{ji}^{(\ell)}$ is the edge descriptor. The message operator applies edge-dependent input and output channel scales around a shared-weight tensor product, followed by equivariant activation. Gated aggregation combines a scalar gate, nonscalar channel gates and relation-specific exponential distance decay. After gate-normalized aggregation, nonscalar magnitudes receive learned invariant rescaling within each angular-order group. The post-convolution block applies an equivariant linear map, activation and channel dropout. Its output is added to the original, pre-normalization hidden state before adaptive normalization. Scalar channels use SiLU, non-scalar channels use invariant norm gates, and dropout masks are shared over the $2l + 1$ components of each irreducible channel.

In Eq. (S14), ℓ is the layer index, $\mathbf{h}_i^{(\ell)}$ the feature vector of node i , $\mathcal{N}(i)$ its incoming neighbors, and $b(i)$ its complex. The unit direction $\hat{\mathbf{r}}_{ji}$, written $\hat{\mathbf{r}}_{ij}$ in Fig. 2, points from sender j to receiver i . Norm denotes equivariant root-mean-square normalization, GateMean the learned gate-normalized neighbor aggregation, Block the post-convolution transformation, and AdaLN adaptive layer normalization. Angular order $l = 0, 1, 2$, distinct from layer depth, has $2l + 1 = 1, 3, 5$ components per channel. Even and odd parity describe behavior under inversion; the evaluated symmetry claim concerns proper joint rotations.

Architecture-diagram notation. In Fig. 2, $c = E_t(t) + E_\sigma(\log \sigma)$ is the conditioning vector called $\boldsymbol{\eta}_b$ here; E_t and E_σ are learned embeddings. Diagram superscript (k) counts layers, whereas subscript f indexes fragments; diagram I_f is the unit-weight geometric inertia $\mathbf{J}_f$ of Eq. (S15), not the mass-weighted guidance inertia. α mixes terminal docking features into the confidence embedding and is initialized to 0.25. h_{in} , h_{conv} and h' denote convolution input, convolution output and transformed features. $\delta_{\text{in/out}}$ are learned radial channel-scale offsets. Distance d_{ij} and learned positive relation-specific scale σ_{type} define distance decay; the latter is distinct from the translation prior σ . RMSNorm is root-mean-square normalization; scalar blocks normalize by scalar RMS and nonscalar blocks by RMS channel norm, with a numerical denominator stabilizer. γ_0, β_0 are scalar affine scale/shift outputs, while $\gamma_{>0}$ controls nonscalar rescaling. The subscript > 0 refers to angular order, not the sign of γ . Its 0.1 tanh factor bounds modulation to 10% about identity. g is the invariant norm-dependent gate, shared over each channel’s components. SiLU is a sigmoid(a); MLP denotes multilayer perceptron and concat denotes concatenation. Channel counts, layer depths, radial features and dropout specify the retained architecture.

Ligand attributes encode element, formal charge, aromaticity, hybridization, valence, chirality, ring membership, and pharmacophore flags. Protein attributes encode residue–atom identity, backbone status, supported-metal identity, and protein-side pharmacophore flags. The distance basis uses 32 Gaussian functions spanning 0–16 Å with width 0.5 Å; unavailable discrete attributes use explicit sentinel categories. Joint rigid motion of all coordinates rotates the predicted fragment translation and angular velocities and leaves invariant outputs unchanged.

S3.2 Atom readout and observable fragment motion

The terminal ligand-atom state is mapped to a polar atom-vector proposal $\mathbf{f}_i$ by an equivariant linear branch plus a zero-initialized self-tensor-product branch. It is not assumed to be an energy gradient.

Table S4: Node types and relation groups in the heterogeneous docking graph. Every directed relation is restricted to one complex in a minibatch.

Component	Construction
Node types	Ligand heavy atoms, rigid-fragment virtual nodes, protein atoms, and residue virtual nodes.
Ligand topology	Retained covalent bonds, cut bonds, cross-cut triangulation pairs, atom–fragment membership, and a complete directed fragment context graph with cut-graph hop distance.
Protein topology	Canonical intra-residue, peptide, and disulfide bonds; atom–residue membership; residue pairs within 10 Å.
Protein–ligand context	Complete residue–fragment context edges and bidirectional protein-atom–ligand-atom contacts within 5 Å, rebuilt at every forward pass.
Edge features	Relation and bond embeddings, 32 Gaussian distance features, optional reference-distance evolution, fragment-local displacement, typed contact flags, endpoint scalar states, and flow conditioning.

For fragment k , with $\mathbf{r}_i = \mathbf{x}_i - \mathbf{T}_k$, the readout is

$$\begin{aligned}
\hat{\mathbf{v}}_k &= \frac{1}{|\mathcal{A}_k|} \sum_{i \in \mathcal{A}_k} \mathbf{f}_i, & \boldsymbol{\mu}_k &= \sum_{i \in \mathcal{A}_k} \mathbf{r}_i \times \mathbf{f}_i, \\
\mathbf{J}_k &= \sum_{i \in \mathcal{A}_k} (\|\mathbf{r}_i\|_2^2 \mathbf{I}_3 - \mathbf{r}_i \mathbf{r}_i^\top), & \hat{\boldsymbol{\omega}}_k &= \mathbf{J}_k^+ \boldsymbol{\mu}_k.
\end{aligned} \tag{S15}$$

The symmetrized inertia matrix is diagonalized in double precision. For multi-atom fragments, eigenmodes not exceeding 1% of the largest eigenvalue are removed; singleton fragments have no observable rotation. The same retained-mode projector $\mathbf{P}_k$ is applied to the angular target.

S3.3 Docking objective

The atom-velocity auxiliary target couples translation and rotation:

$$\hat{\mathbf{u}}_i = \hat{\mathbf{v}}_{g(i)} + \hat{\boldsymbol{\omega}}_{g(i)} \times \mathbf{r}_i, \quad \mathcal{L}_{\text{atom}} = \frac{1}{3N} \sum_{i=1}^N \|\hat{\mathbf{u}}_i - \mathbf{u}_i^*\|_2^2. \tag{S16}$$

For the distance-geometry term, the predicted state is extrapolated from t_b to the endpoint using $(1 - t_b)\hat{\mathbf{v}}_k$ and $(1 - t_b)\hat{\boldsymbol{\omega}}_k$. If $\mathcal{P}_b$ is the set of interfragment atom pairs and ℓ_b their mean squared endpoint-distance error, then

$$\mathcal{L}_{\text{DG}} = \frac{\sum_{b \in \mathcal{B}_+} t_b^2 \ell_b}{\sum_{b \in \mathcal{B}_+} t_b^2}, \quad \mathcal{B}_+ = \{b : |\mathcal{P}_b| > 0\}, \tag{S17}$$

with an exact zero returned when the valid set or total weight is empty. The primary losses are Cartesian-component-averaged translation MSE and observable-mode angular MSE,

$$\mathcal{L}_v = \frac{1}{3K} \sum_k \|\hat{\mathbf{v}}_k - \mathbf{v}_k^*\|_2^2, \quad \mathcal{L}_\omega = \frac{1}{3|\mathcal{K}_{\text{rot}}|} \sum_{k \in \mathcal{K}_{\text{rot}}} \|\hat{\boldsymbol{\omega}}_k - \mathbf{P}_k \boldsymbol{\omega}_k^*\|_2^2. \tag{S18}$$

An empty rotational set contributes zero. Here $\mathcal{K}_{\text{rot}}$ contains fragments with retained rotational modes, $\mathbf{P}_k$ projects onto those modes, and $\mathbf{u}_i^*$ is the atom velocity induced by the target fragment translation and angular velocity. The factors $3N$, $3K$ and $3|\mathcal{K}_{\text{rot}}|$ average squared errors over Cartesian components and valid items; they do not weight complexes by success. In Eq. (S17), t_b is complex b 's flow time, $\mathcal{P}_b$ its interfragment atom-pair set, and ℓ_b its mean squared endpoint-distance error. The t_b^2 weight reduces the contribution of early extrapolations. The coefficients 8, 0.3 and 3 set relative task emphasis in the retained checkpoint's numerical units; their provenance follows the common convention in Sec. S1.1. The released objective is

$$\mathcal{L}_{\text{geom}} = \mathcal{L}_v + 8\mathcal{L}_\omega + 0.3\mathcal{L}_{\text{atom}} + 3\mathcal{L}_{\text{DG}}. \tag{S19}$$

Apart from the explicit t_b^2 factor in $\mathcal{L}_{\text{DG}}$, no loss receives an additional time-dependent weight. All reductions use valid-item counts and distributed count rescaling so that unequal graph sizes approximate a global masked mean.

S4 Confidence Model Details

The confidence model predicts within-bank pose quality for ranking. For each candidate, the fixed docking network is re-evaluated at $t = 1$ with recovered fragment frames and the source prior scale to obtain ligand-atom and fragment features. These features are mixed into a fresh heterogeneous-graph representation through a learned gate initialized to 0.25, followed by four additional equivariant interaction layers. Each layer retains 384 scalar channels, 32 channels of each vector parity and 16 channels of each rank-2 parity, as in the docking trunk; the invariant readout MLPs have hidden width 512. Prior-scale dependence enters the composite score through the docking features; the confidence interaction layers use a zero conditioning vector rather than a separate prior-scale input.

S4.1 Invariant readout

Scalar channels and norms of non-scalar channels form invariant node descriptors. For each ligand atom, the model also uses the nearest protein distance, 32 radial features over 0–10 Å, contact counts within 2, 3.5, and 5 Å, an exponentially weighted contact count, and weighted protein interaction-type flags. Atom heads predict log-transformed displacement and a 2 Å success logit. A pose representation concatenates ligand-contact attention pooling with type-wise mean and maximum pooling over ligand atoms, fragments, protein atoms, and residue nodes. The pose head returns a log-RMSD value $\hat{\ell}_m^{\text{pose}}$ and a success logit. The score used for ranking is

$$\hat{r}_m = \max \left\{ 0, \exp \left[\text{clip}(\hat{\ell}_m^{\text{pose}}, -2, 5) \right] - 1 \right\}. \quad (\text{S20})$$

Lower $\hat{r}_m$ ranks earlier. The clip limits $-2, 5$ act on dimensionless predicted log-RMSD; exp-minus-one reverses the log-one-plus target transformation, and the outer maximum prevents negative reported RMSD. Clipping is a fixed reporting safeguard, not a success threshold. Contact-count radii define local proximity summaries; the exponential count uses weight $\exp[-d/(2 \text{ Å})]$ for atom–protein distance d , and its scale is a fixed feature setting.

S4.2 Targets and objective

Atom targets use fixed-correspondence displacements $d_{mi}^* = \|\mathbf{x}_{mi} - \mathbf{x}_i^*\|_2$, whereas pose targets use symmetry-aware, no-alignment RMSD r_m^* . The pointwise terms are unit-transition Huber losses on $\log(1 + d_{mi}^*)$ and $\log(1 + r_m^*)$, together with atom- and pose-level binary cross entropy for the 2 Å threshold. The pairwise ranking loss applies margin 0.05 to pose pairs whose true log-RMSD gap exceeds 0.3. The success-focused listwise target is proportional to $\text{sigmoid}[(2 \text{ Å} - r_m^*)/(0.2 \text{ Å})]$ within each pose set. With the corresponding losses denoted by their subscripts, the complete objective is

$$\begin{aligned} \mathcal{L}_{\text{conf}} = & 0.2\mathcal{L}_{\text{atom-reg}} + 0.2\mathcal{L}_{\text{atom-BCE}} + 0.3\mathcal{L}_{\text{pose-reg}} + 0.4\mathcal{L}_{\text{pose-BCE}} \\ & + 0.1\mathcal{L}_{\text{rank}} + 1.0\mathcal{L}_{\text{success-listwise}}. \end{aligned} \quad (\text{S21})$$

Empty admissible pair sets contribute zero to $\mathcal{L}_{\text{rank}}$. Crystal anchors may enter confidence training but are excluded from validation model selection. The reg and BCE subscripts denote Huber regression and binary cross entropy, each averaged over its atom or pose targets. The Huber transition 1 and ranking margin/gap 0.05/0.3 are in log-target units. Rank loss averages the hinge penalty over admissible ordered pairs. The success-listwise term is cross entropy between the normalized soft-success target and a softmax over pose success logits; it trains the auxiliary success head, not the deployed predicted-RMSD selector. The 0.2 Å width smooths the target near 2 Å, and the six objective coefficients are fixed relative training weights.

S5 Interaction Energy and Rigid-Fragment Refinement

S5.1 Definition and uses

The guidance subsystem evaluates one scalar function of the ligand coordinates,

$$E_G(\mathbf{x}) = E_{\text{phys}}(\mathbf{x}) + E_{\text{int}}(\mathbf{x}), \quad \mathbf{F}_i = -\frac{\partial E_G}{\partial \mathbf{x}_i}. \quad (\text{S22})$$

Its physical and interaction components are

$$\begin{aligned} E_{\text{phys}} &= E_{\text{bond}} + E_{\text{angle}} + E_{\text{proper}} + E_{\text{chiral}} + E_{\text{planar}} + E_{\text{LJ}} + E_{\text{steric}} + E_{\text{obs}}, \\ E_{\text{int}} &= E_{\text{hyd}} + E_{\text{HB}} + E_{\text{chg}} + E_{\pi} + E_{\text{cat-}\pi} + E_{\text{XB}} + E_{\text{metal}}. \end{aligned} \quad (\text{S23})$$

The receptor is fixed, and distances are expressed in Angstrom. Individual terms use nominal kcal/mol units in a geometric potential. Its gradient is used for coordinate diagnostics, optional inference-time drift, and post-sampling refinement. Inference guidance changes the numerical vector field without modifying the learned network parameters. This use of an auxiliary inference-time field is related to guidance formulations for flow matching, while the energy, normalization, and fragment projection below are specific to EFF-Dock [10]. The following subsections define the fixed parameters and interaction types.

S5.2 Shared smooth functions and site saturation

Let $\bar{u} = \min(1, \max(0, u))$. For inner and outer cutoffs $a < b$, distance r , and squared distance $z = r^2$, the common quintic polynomial and decreasing switches are

$$Q_5(u) = 6\bar{u}^5 - 15\bar{u}^4 + 10\bar{u}^3, \quad (\text{S24})$$

$$s_{\text{cut}}(r; a, b) = Q_5\left(\frac{b-r}{b-a}\right), \quad (\text{S25})$$

$$s_{\text{cut},2}(z; a, b) = Q_5\left(\frac{b^2-z}{b^2-a^2}\right). \quad (\text{S26})$$

Thus $s_{\text{cut}} = 1$ at and below a , $s_{\text{cut}} = 0$ at and above b , and its value, first derivative, and second derivative are continuous at both boundaries. The squared-distance form $s_{\text{cut},2}$ is used when smoothness at exact coordinate overlap is required.

Contact terms use a soft union. For weights $w_{ij} \in [0, 1]$,

$$\text{softOR}_j(w_{ij}) = 1 - \prod_j [1 - (1 - \delta)w_{ij}], \quad \delta = 10^{-7}. \quad (\text{S27})$$

Clamping precedes the logarithmic implementation of this expression. The scalar u is a switch argument, and $\bar{u}$ clips it to $[0, 1]$. The coefficients 6, -15 and 10 give the quintic interpolation with zero first and second endpoint derivatives. The floor $\delta = 10^{-7}$ keeps the logarithmic soft union finite at unit pair weight. Site occupancy saturates rather than rewarding unlimited contacts with one site.

S5.3 Static typing and receptor admission

Ligand interaction typing is evaluated once on the sanitized heavy-atom graph with RDKit [1]; energy and force evaluation does not call RDKit. Protein typing uses canonical residue and atom-name tables with expected heavy-atom connectivity. PDB bond orders are not inferred from distance. Table S5 summarizes the interaction types.

All external refinements and guidance controls use the geometry-only receptor policy: unresolved fixed atoms contribute bounded repulsion without hydrogen-bond or charge attraction. Non-finite input coordinates are rejected.

Table S5: Static interaction typing used by the guidance energy. Incomplete or ambiguous attractive sites are excluded.

Family	Admission rule
H-bond ligand	SMARTS patterns for neutral/cationic N–H, aromatic [nH], hydroxyl donors, and N/O acceptors; amide-like acceptor exclusions are explicit. Only N and O participate.
H-bond protein	Complete canonical N/O sites with specified linear, trigonal, or tetrahedral heavy-atom geometry. Plain HIS side-chain nitrogens, termini, chain gaps, degree mismatches, and missing neighbors are excluded. HID/HIE/HIP retain their input state.
Hydrophobe	Carbon only. Ligand carbons adjacent to N/O/F are excluded by SMARTS; protein carbons use a curated atom-name table. S and halogens are not hydrophobes in this profile.
Formal charge	Ligand carboxylate, guanidinium, and amidinium groups are represented by one equally weighted resonance-group site; remaining nonzero formal charges are atom centered. Protein sites are ARG/LYS +1, ASP/GLU –1, and explicitly named HIP +1.
Aromatic systems	Complete aromatic ring systems define centroids and normals for pi stacking and cation–pi geometry.
Halogen bond	Ligand halogen-donor patterns and receptor acceptor templates provide distance and angular gates.
Metal coordination	Recognized receptor-metal profiles specify admitted ligand donors, coordination geometry, capacity, and repulsion-only fallbacks.

S5.4 Physical energy at fragment interfaces

Because every fragment is rigid, only geometry spanning distinct fragments can change. Let $\mathcal{B}$ be the cut bonds, $\mathcal{H}$ the cross-fragment angles, $\mathcal{D}$ the retained proper-torsion quadruplets, and $\mathcal{I}_\chi$ and $\mathcal{I}_p$ the chiral and planar improper sets. For proper term $p \in \mathcal{D}$, w_p is its normalized multiplicity weight, ϕ_p its dihedral angle, and (n_p, δ_p) its periodicity and phase. The function $\text{wrap}(\psi)$ returns the representative of angle ψ in $(-\pi, \pi]$.

$$E_{\text{bond}} = \sum_{b \in \mathcal{B}} \frac{k_b}{2} (r_b - r_b^{\text{ref}})^2, \quad (\text{S28})$$

$$E_{\text{angle}} = \sum_{a \in \mathcal{H}} \frac{k_a}{2} (\theta_a - \theta_a^{\text{ref}})^2, \quad (\text{S29})$$

$$E_{\text{proper}} = \sum_{p \in \mathcal{D}} k_p^\phi w_p [1 + \cos(n_p \phi_p - \delta_p)], \quad (\text{S30})$$

$$E_{\text{chiral}} = \sum_{\iota \in \mathcal{I}_\chi} \frac{k_\chi}{2} \text{wrap}(\phi_\iota - \phi_\iota^{\text{ref}})^2, \quad (\text{S31})$$

$$E_{\text{planar}} = \sum_{\iota \in \mathcal{I}_p} \frac{k_{\text{planar}}}{2} [1 - \cos(2\phi_\iota)]. \quad (\text{S32})$$

In primary measurements, the bond, angle and chiral reference values are obtained from the chemical reference conformer rebuilt using the recorded generation preparation and conformer seed: zero for Astex and PoseBusters, and the per-complex sampling seed for PhiBench-derived, FoldBench and OpenBind. Refinement starts from the actual saved candidate and fixes each fragment’s internal coordinates to that candidate. The regenerated conformer parameterizes cut-interface restraints rather than replacing the generated pose.

Refinement and confidence use matching conformer templates and fragment partitions. Compatibility with generated poses was checked by independent proper fragment fits, with whole-ligand heavy-atom RMSD at most 10^{-3} Å. Original generation-template hashes were unavailable, so this geometric check does not establish identity of the fragment body-axis convention. Previously refined PhiBench-

derived, FoldBench and OpenBind coordinates and energy references were verified unchanged and reused without rerunning refinement; rescoring used the evaluated frame convention.

The proper term is independent of its starting torsion, and the weights around each cut bond sum to one. Conjugated or aromatic interfaces use $n_p = 2, \delta_p = \pi$; other cut bonds use $n_p = 3, \delta_p = 0$. Chiral improper restraints are active; the separate chemical-constraint auxiliary channel has zero weight. The coefficients are given in Table S6. Here r_b , θ_a and ϕ_p are bond length, bond angle and proper dihedral; ref denotes the specified reference-conformer value. The k coefficients are restraint strengths, not learned parameters. Twofold and threefold profiles specify generic periodic preferences for the stated interface classes. Their coefficients and the steric-barrier settings are fixed custom geometric parameters; only the stated element-level UFF constants and radius tables are taken from the cited physical parameter sources.

Let $\mathcal{N}_{\text{LJ}}$ be the admitted ligand–ligand and protein–ligand nonbonded pairs after topological exclusions. For $(i, j) \in \mathcal{N}_{\text{LJ}}$, $r_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$, s_{ij} is the topological scale, and x_i, D_i are the UFF size and well-depth parameters. The saved-pose refiner uses soft-core length $a_{\text{LJ}} = 0.75 \text{ \AA}$:

$$\tilde{r}_{ij} = (r_{ij}^2 + a_{\text{LJ}}^2)^{1/2}, \quad x_{ij} = (x_i x_j)^{1/2}, \quad D_{ij} = (D_i D_j)^{1/2}, \quad (\text{S33})$$

$$E_{ij}^{\text{LJ}} = s_{ij} s_{\text{cut}}(r_{ij}; 6, 8) D_{ij} \left[\left(\frac{x_{ij}}{\tilde{r}_{ij}} \right)^{12} - 2 \left(\frac{x_{ij}}{\tilde{r}_{ij}} \right)^6 \right], \quad E_{\text{LJ}} = \sum_{(i,j) \in \mathcal{N}_{\text{LJ}}} E_{ij}^{\text{LJ}}. \quad (\text{S34})$$

The element-level form and x, D constants follow UFF [11], but the complete guidance energy is not UFF, MMFF, GAFF, OpenFF, or AMBER. Ligand 1–2 and 1–3 pairs are excluded, 1–4 pairs use $s_{ij} = 0.5$, and all remaining ligand pairs must belong to different fragments.

The default physical profile adds a compact heavy-atom barrier for every included protein–ligand pair, where L and P denote the ligand and admitted receptor heavy-atom sets. With versioned van der Waals radii R_i^{vdW} and R_j^{vdW} , based on standard element radii where available [12], define

$$d_{ij}^{\text{safe}} = \rho (R_i^{\text{vdW}} + R_j^{\text{vdW}}), \quad (\text{S35})$$

$$p_{ij} = \tau \text{ softplus} \left(\frac{d_{ij}^{\text{safe}} - r_{ij}}{\tau} \right), \quad (\text{S36})$$

$$E_{\text{steric}} = \sum_{i \in L} \sum_{j \in P} \frac{k_{\text{steric}}}{2} p_{ij}^2 s_{\text{cut}}(r_{ij}; d_{ij}^{\text{safe}}, d_{ij}^{\text{safe}} + \Delta_{\text{cut}}). \quad (\text{S37})$$

The defaults are $\rho = 0.8$, $k_{\text{steric}} = 20 \text{ kcal mol}^{-1} \text{ \AA}^{-2}$, $\tau = 0.1 \text{ \AA}$, and $\Delta_{\text{cut}} = 0.5 \text{ \AA}$. The barrier has compact support and joins zero with continuous first and second radial derivatives at its outer cutoff. It is smooth away from exact atom coincidence. The barrier discourages receptor overlap during coordinate optimization.

Under the geometry-only receptor policy, unresolved receptor atoms contribute repulsion without attractive interaction typing. Obstacles with admitted element-level UFF parameters use only the repulsive branch of Eq. (S34). For remaining generic obstacles $j \in \mathcal{O}$,

$$a_{ij} = \left(\frac{1.7 \text{ \AA}}{\sqrt{r_{ij}^2 + (0.5 \text{ \AA})^2}} \right)^{12}, \quad (\text{S38})$$

$$E_{\text{obs, generic}} = \sum_{i \in L, j \in \mathcal{O}} (10 \text{ kcal mol}^{-1}) \frac{a_{ij}}{1 + a_{ij}} s_{\text{cut}}(r_{ij}; 2.5 \text{ \AA}, 3.0 \text{ \AA}). \quad (\text{S39})$$

The total obstacle term E_{obs} is the sum of the typed and generic repulsive terms; neither provides a hydrogen-bond, charge or dispersion attraction. In the barrier, ρ scales the radius sum, τ controls smooth penetration, and Δ_{cut} is the support margin. Generic-obstacle length $1.7 \text{ \AA}$, soft-core length $0.5 \text{ \AA}$, amplitude 10 kcal/mol , exponent 12 and 2.5 – $3.0 \text{ \AA}$ switch define the bounded repulsive fallback. In the LJ term, 12 and 6 are repulsive and attractive exponents; 1–2, 1–3 and 1–4 refer to atoms separated by one, two and three bonds. Soft cores regularize close approaches, and outer cutoffs limit interaction neighborhoods.

Table S6: Default physical parameters of the guidance energy.

Parameter	Value	Role
Bond / angle coefficient	160 / 35	Harmonic restraint across a cut interface ($\text{kcal mol}^{-1} \text{\AA}^{-2}$ for bonds; $\text{kcal mol}^{-1} \text{rad}^{-2}$ for angles)
Proper / planar coefficient	0.5 / 12	Periodic torsion / planar restraint (kcal/mol)
Chiral coefficient	12	Harmonic improper restraint ($\text{kcal mol}^{-1} \text{rad}^{-2}$)
LJ 1–4 scale	0.5	Intraligand topological exclusion
LJ switch / cutoff	6/8 $\text{\AA}$	C^2 radial truncation
Steric radius scale / coefficient	0.8/20	Protein–ligand compact overlap barrier (coefficient in $\text{kcal mol}^{-1} \text{\AA}^{-2}$)
Steric softplus / cutoff margin	0.1/0.5 $\text{\AA}$	Smooth penetration and compact support

S5.5 Hydrophobic and directional hydrogen-bond terms

Let $\mathcal{H}_L$ and $\mathcal{H}_P$ be the typed ligand and protein carbon hydrophobe sites. For $i \in \mathcal{H}_L$ and $j \in \mathcal{H}_P$,

$$w_{ij}^{\text{hyd}} = s_{\text{cut}}(r_{ij}; 3.5, 4.5), \quad E_{\text{hyd}} = -0.25 \sum_i \text{softOR}_j(w_{ij}^{\text{hyd}}). \quad (\text{S40})$$

The contact window is centered around the structural criterion used by PLIP [13].

Explicit hydrogens and lone pairs are not required. Bond lengths are numerical values in Angstrom; the normalization floor is $\epsilon = 10^{-8}$. The same numerical floor is also used for dimensionless direction-vector norms. For typed heavy-atom site i , heavy neighbor k , and bond length l_{ik} , define

$$q_b(l) = Q_5 \left(\frac{l - 0.10}{0.50 - 0.10} \right), \quad (\text{S41})$$

$$\tilde{\mathbf{v}}_i = \sum_{k \in \mathcal{N}(i)} q_b(l_{ik}) \frac{\mathbf{x}_i - \mathbf{x}_k}{\max(l_{ik}, \epsilon)}, \quad (\text{S42})$$

$$\mathbf{v}_i = \frac{\tilde{\mathbf{v}}_i}{\max(\|\tilde{\mathbf{v}}_i\|, \epsilon)}, \quad (\text{S43})$$

$$q_i^{\text{axis}} = Q_5 \left(\frac{\|\tilde{\mathbf{v}}_i\| - 0.10}{0.40 - 0.10} \right) \prod_{k \in \mathcal{N}(i)} q_b(l_{ik}). \quad (\text{S44})$$

RDKit hybridization and heavy degree select the missing-valence target cosine t_i : linear degree one gives 1; trigonal degree one/two gives 1/2, 1; and tetrahedral degree one/two/three gives 1/3, 1/ $\sqrt{3}$, 1. Canonical protein sites use the geometry and expected degree specified by the residue templates.

Let $C(c; t, \Delta)$ be a two-sided Q_5 cone gate centered on target cosine t , compactly supported within angular half-width Δ . For donor D , acceptor A , and $\hat{\mathbf{r}}_{DA} = (\mathbf{x}_A - \mathbf{x}_D)/r_{DA}$,

$$R_{\text{HB}}(r) = \exp \left[-\frac{1}{2} \left(\frac{r - 2.9}{0.35} \right)^2 \right] s_{\text{cut}}(r; 3.5, 4.1), \quad (\text{S45})$$

$$w_{DA}^{\text{HB}} = R_{\text{HB}}(r_{DA}) C(\mathbf{v}_D^\top \hat{\mathbf{r}}_{DA}; t_D, 60^\circ) C(-\mathbf{v}_A^\top \hat{\mathbf{r}}_{DA}; t_A, 45^\circ) q_D^{\text{axis}} q_A^{\text{axis}}, \quad (\text{S46})$$

$$E_{\text{HB}} = -1.0 \sum_D \text{softOR}_A(w_{DA}^{\text{HB}}). \quad (\text{S47})$$

Here t_D and t_A are the target cosines assigned to the donor and acceptor by the preceding valence rule. In Eq. (S44), q_b suppresses collapsed bonds and q_i^{axis} suppresses poorly defined site directions; the

0.10/0.50 bond-length limits use Angstrom, whereas 0.10/0.40 axis-norm limits are dimensionless. $\tilde{\mathbf{v}}_i$ is the weighted missing-valence direction sum and $\mathbf{v}_i$ its normalized direction, distinct from a fragment velocity. Ideal linear, trigonal and tetrahedral geometry determines the target cosines. The hydrogen-bond distance peak 2.9 Å, width 0.35 Å, 3.5–4.1 Å switch and 60°/45° donor/acceptor cone widths are fixed geometric gates; their values are not inferred from the evaluated poses. Ligand-donor/protein-acceptor and protein-donor/ligand-acceptor directions are both evaluated. If the axis norm is at most ϵ , its direction and interaction weight are set to zero.

S5.6 Screened formal-charge groups

Each charge site a has integer charge Q_a and normalized nonoverlapping membership W_{ai} , so $\mathbf{R}_a = \sum_i W_{ai} \mathbf{x}_i$. Charge conservation between sites and the sanitized ligand is checked at construction. Let $\mathcal{Q}_L$ and $\mathcal{Q}_P$ be the ligand and protein formal-charge sites. For $(a, b) \in \mathcal{Q}_L \times \mathcal{Q}_P$,

$$d_{ab}^2 = \|\mathbf{R}_a - \mathbf{R}_b\|^2, \quad \rho_{ab} = (d_{ab}^2 + 0.75^2)^{1/2}, \quad (\text{S48})$$

$$E_{ab}^{\text{chg}} = \frac{332.06371}{78.5} Q_a Q_b \frac{\exp(-0.127 \rho_{ab})}{\rho_{ab}} Q_5 \left(\frac{10^2 - d_{ab}^2}{10^2 - 8^2} \right), \quad (\text{S49})$$

$$E_{\text{chg}} = \sum_{(a,b) \in \mathcal{Q}_L \times \mathcal{Q}_P} E_{ab}^{\text{chg}}. \quad (\text{S50})$$

Opposite charges are attractive and equal charges repulsive; signed pair energies are summed directly without saturation. Q_a is charge in units of the elementary charge, W_{ai} normalized site membership, and $\mathbf{R}_a$ a site centroid rather than a fragment rotation. d_{ab} is centroid separation and ρ_{ab} its soft-core distance. The prefactor 332.06371 converts elementary-charge pairs and Angstrom to nominal kcal/mol; 78.5 is the relative dielectric and 0.127 Å^{-1} the screening wavenumber. The 0.75 Å soft core and 8–10 Å switch regularize and truncate the pair interaction; distances substituted in the displayed expression are numerical Angstrom values. The Debye–Hückel-inspired screening constant represents 0.15 M monovalent ionic strength, 298.15 K, and bulk-water $\epsilon_r = 78.5$ [14]. The term omits partial charge, protonation inference, heterogeneous dielectric, desolvation, polarization, solvent/counterion reorganization, and receptor relaxation.

S5.7 Aromatic, halogen-bond, and metal terms

The geometric motifs used in this subsection are established protein-structure interaction classes: aromatic stacking, cation–pi contacts, directional halogen bonds, and metal coordination [15, 16, 17, 18, 19, 20, 21].

For typed site sets A and B and clipped pair weights $w_{ij}^X \in [0, 1]$, define one-sided and symmetric saturated occupancies by

$$\text{Occ}_{A \rightarrow B}(w^X) = \sum_{i \in A} \text{softOR}_{j \in B}(w_{ij}^X), \quad (\text{S51})$$

$$\text{SymOcc}_{A,B}(w^X) = \frac{1}{2} [\text{Occ}_{A \rightarrow B}(w^X) + \text{Occ}_{B \rightarrow A}((w^X)^\top)]. \quad (\text{S52})$$

The weight w_{ij}^X is the product of the versioned distance, angular, donor-cone, aromatic-offset, or coordination-profile gates for interaction family X ; the static site definitions are given in Table S5. Let $\mathcal{R}_L, \mathcal{R}_P$ denote ligand and protein aromatic systems, $\mathcal{Q}_L^+, \mathcal{Q}_P^+$ positive-charge sites, $\mathcal{X}_L$ ligand halogen donors, and $\mathcal{A}_P^X$ receptor halogen acceptors. The aromatic and halogen contributions are

$$E_\pi = -0.25 \text{SymOcc}_{\mathcal{R}_L, \mathcal{R}_P}(w^\pi), \quad (\text{S53})$$

$$E_{\text{cat-}\pi} = -0.5 \left[\text{SymOcc}_{\mathcal{R}_L, \mathcal{Q}_P^+}(w^{\text{cat-}\pi}) + \text{SymOcc}_{\mathcal{Q}_L^+, \mathcal{R}_P}(w^{\text{cat-}\pi}) \right], \quad (\text{S54})$$

$$E_{\text{XB}} = -0.5 \text{Occ}_{\mathcal{X}_L \rightarrow \mathcal{A}_P^X}(w^{\text{XB}}). \quad (\text{S55})$$

For admitted metal profile h , let E_h^{pair} , E_h^{over} , E_h^{slot} , and E_h^{non} denote its donor-pair attraction, over-coordination penalty, coordination-slot penalty, and non-donor repulsion. The total metal term is

$$E_{\text{metal}} = \sum_{h \in \mathcal{M}} \left(E_h^{\text{pair}} + E_h^{\text{over}} + E_h^{\text{slot}} + E_h^{\text{non}} \right), \quad (\text{S56})$$

where $\mathcal{M}$ is the set of recognized receptor-metal profiles. Unsupported profiles have no attractive pair or slot contribution and remain repulsion only. The low-level gate thresholds and metal-profile capacities are versioned implementation parameters rather than learned quantities. The sets $\mathcal{Q}_L^+$, $\mathcal{Q}_P^+$ contain the positive-charge sites from $\mathcal{Q}_L$, $\mathcal{Q}_P$. X indexes an interaction family and h a metal profile. Occ and SymOcc sum one-sided and symmetric site occupancies; their factor 1/2 averages the two directions. Negative principal coefficients favor admitted contacts, and all displayed attraction amplitudes use nominal kcal/mol. These gates and coefficients specify the fixed interaction profile.

Table S7: Active interaction terms in the guidance energy.

Term	Aggregation and principal coefficient
Hydrophobic	Ligand-site soft occupancy; $\varepsilon_{\text{hyd}} = 0.25$.
Hydrogen bond	Both donor-acceptor directions; $\varepsilon_{\text{HB}} = 1.0$.
Formal charge	Signed screened Coulomb pair sum with dielectric 78.5.
Pi stacking	Symmetric ring-system occupancy; $\varepsilon_{\pi} = 0.25$.
Cation-pi	Symmetric occupancy in both directions; $\varepsilon_{\text{cat-}\pi} = 0.5$.
Halogen bond	Ligand-donor/receptor-acceptor occupancy; $\varepsilon_{\text{XB}} = 0.5$.
Metal coordination	Profile-specific pair, overcoordination, slot, and non-donor contributions.

S5.8 Mass-inertia projection

For fragment k , let m_i be the mass of atom i , m_k^{frag} the fragment mass, $\mathbf{c}_k$ the center of mass, and $\mathbf{T}_k$ the geometric-centroid state origin.

$$m_k^{\text{frag}} = \sum_{i \in \mathcal{A}_k} m_i, \quad \mathbf{c}_k = (m_k^{\text{frag}})^{-1} \sum_{i \in \mathcal{A}_k} m_i \mathbf{x}_i, \quad (\text{S57})$$

$$\mathbf{F}_k = \sum_{i \in \mathcal{A}_k} \mathbf{F}_i, \quad \boldsymbol{\tau}_k = \sum_{i \in \mathcal{A}_k} (\mathbf{x}_i - \mathbf{c}_k) \times \mathbf{F}_i, \quad (\text{S58})$$

$$\mathbf{I}_k = \sum_{i \in \mathcal{A}_k} m_i [\|\mathbf{x}_i - \mathbf{c}_k\|^2 \mathbf{I}_3 - (\mathbf{x}_i - \mathbf{c}_k)(\mathbf{x}_i - \mathbf{c}_k)^\top], \quad (\text{S59})$$

$$\mathbf{d}_k^\omega = \mathbf{I}_k^+ \boldsymbol{\tau}_k, \quad \mathbf{d}_k^T = \mathbf{F}_k / m_k^{\text{frag}} + \mathbf{d}_k^\omega \times (\mathbf{T}_k - \mathbf{c}_k). \quad (\text{S60})$$

The truncated pseudoinverse is evaluated in double precision after symmetrizing $\mathbf{I}_k$. Eigenmodes not exceeding 1% of the largest eigenvalue are removed. This mass-inertia-preconditioned projection is distinct from the learned model’s unit-weight atom-vector readout. Atomic masses use the frozen element table in atomic mass units. $\boldsymbol{\tau}_k$ is torque about the center of mass, while $\mathbf{d}_k^T, \mathbf{d}_k^\omega$ are preconditioned search directions, not physical velocities or a molecular-dynamics acceleration. It is recorded without coordinate updates, converted to a normalized inference-time drift, or used by the post-sampling refiner described below.

S5.9 Normalized inference-time guidance drift

For fragment fields $(\mathbf{v}, \boldsymbol{\omega})$, define the induced ligand-atom velocity and its root-mean-square magnitude by

$$\mathbf{U}_i(\mathbf{v}, \boldsymbol{\omega}) = \mathbf{v}_{g(i)} + \boldsymbol{\omega}_{g(i)} \times (\mathbf{x}_i - \mathbf{T}_{g(i)}), \quad (\text{S61})$$

$$\text{RMS}(\mathbf{U}) = \left[\frac{1}{N} \sum_{i=1}^N \|\mathbf{U}_i\|_2^2 \right]^{1/2}. \quad (\text{S62})$$

For integration interval j , with $\Delta t_j = t_{j+1} - t_j$, the interval-averaged late-time ramp is

$$\bar{r}_j = \frac{1}{\Delta t_j} \int_{t_j}^{t_{j+1}} \max(0, 2t - 1) dt. \quad (\text{S63})$$

Let $(\mathbf{d}^T, \mathbf{d}^\omega)$ be the raw mass-inertia directions from Eq. (S60). Before capping, the added drift is

$$\beta_j = \eta \bar{r}_j \frac{\text{RMS}[\mathbf{U}(\hat{\mathbf{v}}, \hat{\boldsymbol{\omega}})]}{\text{RMS}[\mathbf{U}(\mathbf{d}^T, \mathbf{d}^\omega)]}, \quad (\text{S64})$$

$$(\Delta \mathbf{v}_k, \Delta \boldsymbol{\omega}_k) = \beta_j (\mathbf{d}_k^T, \mathbf{d}_k^\omega), \quad (\text{S65})$$

$$\tilde{\mathbf{v}}_k = \hat{\mathbf{v}}_k + \Delta \mathbf{v}_k, \quad \tilde{\boldsymbol{\omega}}_k = \hat{\boldsymbol{\omega}}_k + \Delta \boldsymbol{\omega}_k. \quad (\text{S66})$$

The drift is set to zero when either RMS is at most 10^{-8} or the projected direction is non-finite. A single nonnegative per-pose rescaling caps $\|\Delta \mathbf{v}_k\|_2$ at $5 \text{ \AA}$, $\|\Delta \boldsymbol{\omega}_k\|_2$ at 5 radians per unit flow time, and the estimated interval atom displacement at $0.25 \text{ \AA}$; atom-force norms are capped at 20 before projection. The guidance controls reported in Fig. S19 use $\eta = 2$, the `geometry_only` receptor policy and no Vina or external force-field term. Both networks use the same orientation convention as the primary evaluation. For each active interval, the LJ soft-core length is evaluated at $t_{\text{mid}} = [\max(t_j, 0.5) + t_{j+1}]/2$. It decreases linearly from 1.5 to $0.75 \text{ \AA}$ over the active half of integration. The ligand coordinates and pose-dependent contacts are rebuilt after each guided SE(3) update. This is an inference-only modification; guidance is not present in the geometry-training objective. Here η is dimensionless pre-cap strength, $\bar{r}_j$ the average ramp over interval j , and β_j the factor matching projected atom-motion magnitude to the learned field. The ramp is zero through $t = 0.5$ and rises to one at $t = 1$. The atom-force cap 20 uses nominal $\text{kcal mol}^{-1} \text{ \AA}^{-1}$; the RMS floor prevents division by a negligible direction. Guidance strength, ramp, soft-core schedule and motion caps are fixed control settings.

S5.10 Bounded post-sampling refinement

Refinement begins from a saved generated candidate and updates only its fragment translations and proper rotations. At iteration r , write $E_r = E_G(\mathbf{x}^{(r)})$. The projected directions propose

$$\mathbf{T}_k^{(r+1)} = \mathbf{T}_k^{(r)} + \alpha_r \bar{\mathbf{d}}_k^T, \quad \mathbf{R}_k^{(r+1)} = \text{Exp}(\alpha_r \bar{\mathbf{d}}_k^\omega) \mathbf{R}_k^{(r)}. \quad (\text{S67})$$

The capped increments $(\bar{\mathbf{d}}^T, \bar{\mathbf{d}}^\omega)$ first limit translation to $0.10 \text{ \AA}$ and rotation to 5° per fragment. Both increments are then rescaled together until the maximum displacement of reconstructed atoms is at most $0.10 \text{ \AA}$, before backtracking is applied. A monotone line search starts from base step 1 and tests $\alpha_r = 0.5^b$ for backtracking index $b = 0, \dots, 12$. For the corresponding trial coordinates $\mathbf{x}_{\text{trial}}$, let $E_{\text{trial}} = E_G(\mathbf{x}_{\text{trial}})$. A finite trial is accepted when

$$E_{\text{trial}} \leq E_r + 10^{-10} \max(1 \text{ kcal mol}^{-1}, |E_r|). \quad (\text{S68})$$

The implemented protocol runs for at most 100 refinement iterations and terminates earlier on convergence or line-search failure, with the termination status recorded. It also terminates after 20 consecutive accepted steps with maximum induced atom displacement below $10^{-5} \text{ \AA}$. Starting at iteration 25, a pose stops after five consecutive accepted updates satisfying

$$E_r - E_{r+1} \leq 0.02 \text{ kcal mol}^{-1} + 10^{-3} \max(1 \text{ kcal mol}^{-1}, |E_r|). \quad (\text{S69})$$

All 100 candidates are refined independently; finite coordinates from early-stopped poses define their final output. Raw coordinates are retained for paired analysis. Refinement index r counts optimization iterations, unrelated to an interatomic distance. α_r is the dimensionless accepted backtracking multiplier; $\mathbf{x}_{\text{trial}}$ is its trial geometry. The 13 trial multipliers from 0.5^0 to 0.5^{12} reduce the proposed move until acceptance. The 10^{-10} allowance accommodates energy roundoff; displacement, plateau and iteration limits bound movement and numerical work. Refinement is rigid-fragment SE(3) descent on E_G and does not call the docking network, Vina, or another minimizer.

S5.11 Inputs and numerical verification

The guidance calculation uses receptor coordinates and atom identities, the ligand graph and chemical state, the protocol-specific chemical reference conformer, the supplied pocket and current ligand coordinates. Reference coordinates are used only for complete atom mapping and retrospective RMSD, not for energy, gradients, step acceptance or termination. It does not use the crystal pose, RMSD, benchmark identity, PoseBusters outcome, confidence score, or ranking label. The guidance receptor

shell is separate from the docking network’s 10 Å pocket crop. It contains receptor atoms within 18 Å of the pocket center. Because the largest interaction cutoff is 10 Å, ligand atoms more than 8 Å from the pocket center can have truncated receptor neighborhoods; these cases are recorded per pose.

Numerical verification covered finite-difference force signs, rigid-motion energy invariance and force/projection covariance, topology exclusions, 1–4 scaling, chirality, periodicity, cutoffs, degenerate geometries and formal-charge conservation. These checks assess numerical consistency rather than docking accuracy or component contribution.

S6 Data Curation and Training

S6.1 Curation filters and missing-value policy

Table S8 gives the ordered PLINDER curation procedure [5]. Counts are the numbers of records remaining after each step. Missing resolution, real-space correlation coefficient (RSCC), real-space R factor (RSR), protein-volume overlap, internal-energy, crystal-contact, and interface values follow the stated lenient filtering rule and are not statistically imputed.

Table S8: Ordered PLINDER 2024-06/v2 curation funnel.

Step	Records	Survivor criterion applied at this step
0	1,357,906	Raw full PLINDER plindex
1	467,765	Remove ion, artifact, invalid, and covalent ligand records
2	435,113	At least 3 protein–ligand interactions
3	422,172	Resolution at most 3.5 Å, or missing
4	394,791	Resolved / total ligand heavy atoms at least 0.90
5	386,165	RSCC at least 0.7, or missing
6	381,874	PoseBusters protein-volume overlap below 0.1, or missing
7	359,762	PoseBusters internal-energy pass or cofactor exemption; missing values pass the lenient branch
8	346,582	RSR at most 0.3, or missing
9	336,565	Crystal-contact atom fraction below 0.3, or missing
10	329,770	No missing protein–ligand interface residue; missing count passes the lenient branch
11	326,016	At most 120 ligand heavy atoms
12	322,244	At most 30 ligand rotatable bonds
13	107,443	Deterministic representative per (PDB entry, canonical SMILES)
14	50,297	Deterministic representative per (pocket70 community, canonical SMILES)

Deterministic representative ranking prefers biological assembly 1, followed by more interactions, larger RSCC, smaller RSR, smaller resolution, and stable system and ligand-chain identifiers. Pre-processing succeeded for 50,213 of the 50,297 selected records. The remaining 84 failed because of missing inputs, parsing, featurization or fragment decomposition. The resolved-atom fraction divides observed ligand heavy atoms by the total; RSCC measures real-space model–density correlation and RSR a real-space density residual. Resolution, resolved-atom fraction, RSCC and RSR thresholds screen experimental completeness and fit; interaction/interface and crystal-contact filters screen usable receptor–ligand context. Size and fragmentation limits bound accepted input complexity. These fixed curation rules define the training inventory separately from prediction-success criteria. The `pocket70` label is the supplied PLINDER `pocket_fident__70__community` assignment: a community of its pocket-sequence-identity similarity graph thresholded at 70% [5]. It is a graph-community label, not a guarantee of 70% pairwise identity between every member, nor the binding-chain identity used in our evaluation.

S6.2 Dataset split

Each sample is identified by its PLINDER system ID and ligand-instance chain. The released docking and confidence pipelines start from the same held-out identity split but apply different input filters. Table S9 reports the resulting membership.

Table S9: Docking and confidence training-set membership.

Quantity	Docking	Confidence
Pre-filter training identities	47,310	47,310
Included training samples	47,277	43,092
Excluded training identities	33	4,218
Validation samples	1,076	1,035
Shared training IDs	43,067	
Docking-only / confidence-only IDs	4,210	25
Docking unique systems / PDB IDs	45,441 / 43,392	—

Model-specific input requirements produce distinct training memberships; neither set contains the other. Relatedness analyses use the 47,277-sample docking reference, with the 25 confidence-only samples examined separately.

External exclusion required both an exact ligand match and membership in a benchmark-associated pocket community, removing 904 samples across 217 communities. It did not independently exclude recurring ligands or homologous pockets; training relatedness was measured separately.

S6.3 Confidence-pose dataset

For each eligible training complex, confidence training uses 32 raw production poses, the corresponding 32 deterministically refined poses, and one mapped crystal anchor. The loader draws at most 64 poses per complex and limits the pose-node product to 64,000, with an eight-pose safeguard for exceptionally large graphs. Validation uses up to 100 candidates per complex and excludes the crystal anchor from model selection. Pose-level targets use symmetry-aware, no-alignment RDKit `CalcRMS`; atom-displacement targets retain fixed atom correspondence because a unique symmetry-equivalent per-atom displacement is not defined.

S6.4 Training and checkpoint selection

The evaluated docking model has three sequential training stages (Table S10). The geometry stage was initialized from the terminal base model, and the early-time-replay stage from the geometry checkpoint selected by median internal-validation rollout RMSD. Optimizer and scheduler states were reset between stages; model and available EMA states were retained. The final 50,000-update stage used the time distribution in Eq. (S9) and objective in Eq. (S19). The continuation changed the optimizer, learning rate, batch size, time distribution and auxiliary-loss weights together. Atom-velocity and distance-geometry weights were 0.5/4 in the geometry-stage configuration and 0.3/3 in the final stage.

Table S10: Docking-model training stages. Update counts are local to each stage; model and EMA states continue across stages.

Stage	Updates; global batch	Optimizer peak learning rates	Schedule and retained state
Base model	200,000; 256	Muon 0.02; AdamW 3×10^{-4}	Warmup 4,000; plateau through 100,000; cosine decay to 5% of peak; terminal state initializes the next stage.
Geometry fine-tuning	100,000; 256	Muon 0.01; AdamW 1.5×10^{-4}	Warmup 2,000; plateau through 50,000; cosine decay to 5% of peak; minimum validation-rollout median RMSD selects the terminal 100,000-update checkpoint.
Early-time replay	50,000; 64	AdamW 2×10^{-5}	Warmup 1,000; plateau through 40,000; cosine decay to 2×10^{-6} ; prespecified terminal EMA evaluated.

The confidence model was also trained in stages (Table S11). Its initial checkpoint was selected by the success head, whereas both later internal evaluations rank by predicted RMSD. The terminal

state of the symmetry-aware adaptation was used to initialize raw-and-refined training, rather than that adaptation’s best intermediate checkpoint. The final run evaluated up to 100 candidates for each of 1,035 validation complexes and selected the 70,000-update checkpoint.

Table S11: Confidence-model initialization and selection. Learning rates list Muon parameters first and the remaining AdamW parameters second; update counts are local to each continuation. All selection sets are internal validation sets.

Stage	Training inventory; global batch	Peak learning rates	Schedule and checkpoint use
Initial ranker	4,096; 4	0.02; 3×10^{-4}	50,000-update run, 1,000 warmup and final-half cosine decay to 5%; update 42,500 selected by success-head accuracy on 256 validation complexes.
Symmetry-aware adaptation	43,092; 4	0.002; 3×10^{-5}	50,000 updates, 1,000 warmup and final-half cosine decay to 5%; terminal state initializes the next stage.
Raw-and-refined adaptation	43,092; 2	2×10^{-4} ; 3×10^{-6}	100,000-update run, 2,000 warmup and final-half cosine decay to 5%; update 70,000 selected by predicted-RMSD accuracy on 1,035 validation complexes.

Membership inventories cover the docking and confidence training sets, not a reconstructed union of all ancestral training exposures. Inclusion of every initial-ranker example in the later inventory was not established; the relatedness analysis therefore cannot certify exclusion throughout the complete training history.

Table S12: Final-stage optimization and selection settings for the evaluated docking and confidence weights.

Model	Batch configuration	Optimization	Schedule and selection
Docking model	16 complexes per rank on four GPUs (global 64)	AdamW, peak LR 2×10^{-5} , weight decay 0.01, gradient cap 1.0, EMA 0.999	Linear warmup, constant phase, and cosine decay to 2×10^{-6} ; terminal EMA released
Confidence model	One complex per rank on two GPUs (global 2); seed 45	Muon [22] LR 2×10^{-4} and remaining-parameter LR 3×10^{-6} ; weight decay 0.01; gradient cap 1.0	2% warmup and cosine cooldown over the final 50% to 5% of peak; selected on internal validation

The confidence architecture uses 32 radial basis functions, $l_{\max} = 2$, a 128-dimensional zero conditioning vector, dropout 0.1, a 5 Å contact cutoff and global contact attention. The confidence training loader caps protein-atom nodes at 1,024; coordinate-based benchmark scoring instead uses all retained nodes in the specified 10 Å residue crop, without this loader cap. One complex forms a batch, and pose sets are stratified before batching. For internal model selection, let $\mathcal{V}_{1035}$ be the fixed set of 1,035 validation complexes. Each complex contributes the highest-ranked candidate and the selection metric

$$S_{\text{val}}^{\text{succ}} = \frac{1}{1035} \sum_{c \in \mathcal{V}_{1035}} \mathbb{I}[r_c, \arg \min_m \hat{r}_{cm} < 2 \text{ Å}] \quad (\text{S70})$$

is maximized, where $\hat{r}_{cm}$ is predicted RMSD and r_{cm} is the symmetry-aware validation RMSD. The checkpoint maximizing this fixed internal-validation metric was released; it produced 622/1,035 (60.10%) top-1 successes on that bank. External benchmark outcomes do not enter this selection. Update counts measure optimizer steps, not epochs. Global batch counts complexes across all distributed processes; rank in the batch configuration means process rank, not pose rank. Learning rate sets optimizer step scale, weight decay shrinks parameters, and the gradient cap limits global gradient

norm. EMA means exponential moving average; decay 0.999 weights the previous parameter average by 0.999 and the current parameters by 0.001. Warmup, plateau and cooldown numbers specify the recorded learning-rate schedule. Training seeds identify random streams, and loader node/pose caps bound batch size and memory. Settings and stage lengths record the retained runs; the stated internal-validation criterion selects the confidence checkpoint.

S7 Evaluation Details

S7.1 Evaluation cohorts and protocol

The cohort sources are PLINDER, PoseBusters v2, Astex Diverse Set, the PhysDock/PhiBench-derived release, FoldBench, and OpenBind [5, 23, 24, 25, 26, 27]. PoseBusters v2 denotes the 308-complex journal-paper subset of the original 428-complex benchmark after crystal-contact exclusions [23, 28]. This benchmark release is distinct from the evaluation-software version, PoseBusters 0.6.5.

Table S13: Evaluation cohorts, construction, and intended interpretation.

Cohort	Complexes	Construction and interpretation
Docking validation set	1,076	Preserved held-out identity list used for docking-validation provenance; 41 entries do not enter the confidence validation set.
PLINDER confidence validation	1,035	Fixed candidate bank used for symmetry-aware confidence validation and checkpoint selection; external cohorts do not select the checkpoint.
PoseBusters v2	308	Repeated-use reference-pocket holo-redocking set emphasizing coordinate validity through the official 27 non-RMSD checks as well as pose accuracy.
Astex Diverse Set	85	Small curated diverse reference-pocket redocking set; useful for interpretable method comparison but repeatedly exposed during development.
PhiBench-derived	206	Locally selected from the released archive using temporal and sequence criteria. Two cases lack retrieved deposition-date metadata, and three references contain imputed heavy-atom coordinates. Identity with the published 206-case cohort has not been established.
FoldBench-Pocket full	558	All released protein–ligand interface rows from 554 assemblies, converted to fixed-holo, crystal-pocket redocking. Four multi-ligand assemblies retain distinct chain and ligand identifiers. Two reference ligands each contain one imputed heavy atom. This is not the native FoldBench cofolding leaderboard.
FoldBench later-release slice	66	Subset of the full 558-complex campaign first released after June 30, 2024; treated as a temporal slice rather than an independent cohort.
OpenBind	925	EV-A71/CVA16 2A-protease complexes retained in the full denominator, including two covalent systems treated as noncovalent approximations. This dense target-family set contains repeated chemotypes and closely related receptors rather than broad target diversity.

PhiBench-derived selection used a deposition-date window of 1 June–31 December 2024 where metadata were available; two archive entries without retrieved dates were retained. Receptor-chain sequences were concatenated in archive order, exact sequence duplicates were reduced, and one deterministic representative was retained per connected component at 99.5% global identity, normalized by the longer sequence length. This construction criterion is distinct from the training-relative binding-chain identities reported below.

Missing reference coordinates were completed from a fixed Chemical Component Dictionary ideal conformer after local rigid alignment to observed atoms; observed coordinates were preserved. One heavy atom was imputed in each of three PhiBench-derived references and two FoldBench references. These five constructed coordinates are not experimentally resolved observations. The affected case

identities accompany the source data, and all cases remain in the primary cohort denominators.

The reported results use reference-pocket holo redocking; apo-receptor docking, blind docking, and sequence-conditioned cofolding are separate tasks. The primary setting uses $M = 100$ candidates, $S = 10$ late-biased Euler intervals with power 3, $\sigma = 2 \text{ \AA}$, a $10 \text{ \AA}$ receptor crop, no center jitter, and no inference-time guidance. These values specify the retained reference-pocket protocol: candidate count controls bank size, interval count controls integration work, prior scale controls initial spatial spread, and crop radius controls receptor context. Seed labels 42, 100042 and 200042 identify the three fixed random streams used to measure sampling variation. All five external cohorts use three inference repeats with fixed trained weights under this common unguided protocol. Primary measurements use the feature injection in Eq. (S1) with the frozen released docking and confidence weights. Every saved candidate is refined independently for at most 100 iterations, and raw and refined banks are rescored separately by the selected confidence model. The primary selected-pose condition removes candidates whose assigned stereocenters mismatch the input ligand before confidence selection; if no candidate passes, it falls back to the unfiltered top-1 pose.

For FoldBench-Pocket, two bridged ligands that fail the seed-specific ETKDGv3 construction use a same-seed fallback in which chirality enforcement is relaxed only during distance-geometry embedding; all SMILES-specified chiral centers must still match the resulting structure. Two reference chains missing one heavy-atom coordinate are repaired by a local Kabsch alignment of the ideal Protein Data Bank Chemical Component Dictionary (CCD) conformer while preserving every observed coordinate [4, 29]. Results are retained for the full 558-interface set and the 556 interfaces with fully observed references. The 66 later-release and 492 complementary interfaces are subdivisions of the same FoldBench campaign.

S7.2 Evaluation scope

The external results describe fixed-holo, reference-pocket redocking rather than blind docking, apo-receptor docking, induced-fit docking, or prospective target discovery. Astex and PoseBusters were inspected repeatedly during development; PhiBench-derived and FoldBench-Pocket are task adaptations; and OpenBind is a dense single-protease-family cohort. OpenBind retains 65 records with at least one source flag: covalency (2), a prepared PoseBusters failure flag (6), a reference PoseBusters failure flag (28), or suspected artefact (39). These categories overlap and are source annotations rather than re-evaluated predictions. All 925 records remain in the primary denominator. Training-set relatedness is therefore reported directly rather than inferred from dataset names or deposition dates.

The experiments evaluate the complete released pipeline and within-bank rank correlation, discrimination, regret and empirical reliability. Component effects and cross-generator calibration require separate controlled studies. Numerical consistency tests in Section S5 assess implementation behavior; the three fixed-weight repeats quantify sampling variation on these prepared cohorts.

S7.3 Benchmark-specific comparisons

Table S14 places published comparator results alongside EFF-Dock under the corresponding selection rules and benchmark-specific endpoints. Each row retains its task, endpoint and denominator; source aggregates and locally evaluated EFF-Dock results are distinguished.

For PhiBench, the pocket-conditioned arm of PhysDock’s Figure 2 and Supplementary Table 2 reports PhysDock and seven supplied-pocket baselines using top-5 of 40 candidates [25]. The reported evaluable denominators range from 180 to 206 and are retained. The source uses protein-aligned ligand RMSD (PAL-RMSD) and reports 18 PoseBusters criteria. EFF-Dock uses its locally selected 206-case PhiBench-derived cohort, fixed-receptor symmetry-aware RMSD and 27 non-RMSD PoseBusters checks. The 40-pose restriction is a saved generation-order prefix, not a new matched-budget run. Identity with the published 206 cases, ligand preparation and the historical atom-mapping/validity implementation are not established. The reported check counts alone do not establish different stringency.

For FoldBench, the [official full-set results](#) provide cofolding references [26]. EFF-Dock preserves all 558 interface identities and is reevaluated with the original OpenStructure 2.8.0 binding-site-superposed ligand RMSD and LDDT-PLI endpoint. Cofolding predicts the receptor as well as the ligand, whereas EFF-Dock receives the holo receptor and reference pocket. The released-group size does not establish identical evaluable denominators for every source model, and training exposure differs.

For OpenBind, the [released structural comparison](#) reports GNINA holo redocking on 802 curated follow-on complexes [27]. EFF-Dock uses the same outcome-blind annotation subset for the benchmark-specific analysis, in addition to its 925-case primary cohort. Preparation, candidate budgets and validity implementations are not fully harmonized. Published rates carry no inferred repeat uncertainty; EFF-Dock entries report mean and sample SD over three fixed-weight inference repeats.

EFF-Dock under benchmark-specific endpoints. Saved candidates are also evaluated under definitions corresponding to the additional benchmark releases. FoldBench retains all 558 interface identities and uses OpenStructure 2.8.0 for binding-site-superposed, symmetry-aware ligand RMSD and LDDT-PLI. The raw and refined primary top-1 poses are evaluated separately; the combined endpoint requires $\text{RMSD} < 2 \text{ \AA}$ and $\text{LDDT-PLI} > 0.8$ on the same pose. For OpenBind, the released metadata select the 802 follow-on cases independently of EFF-Dock outcomes. OpenStructure 2.11.1 provides structural scores; $\text{RMSD} \leq 2 \text{ \AA}$ is combined with all 27 PoseBusters 0.6.5 non-RMSD checks, with an additional same-pose $\text{LDDT-PLI} \geq 0.8$ requirement. Raw and refined top-1, top-5 and top-25 are reported. PhiBench-derived uses fixed-receptor symmetry-aware $\text{RMSD} < 2 \text{ \AA}$ and the 27-check conjunction for refined top-1 and top-5; this does not reproduce the source PAL-RMSD or 18-criterion endpoint.

Candidates retain their saved predicted-RMSD ordering and input-chirality eligibility, with the same all-ineligible fallback. The OpenBind 25-pose and PhiBench-derived 40-pose restrictions select from saved generation-order prefixes of the 100-pose banks; they are not fresh generations at matched computational cost. Unassigned OpenStructure structural scores remain explicit; PoseBusters uses the conservative pass-all policy specified below. Confirmed-success rates retain every complex, while conditional rates distinguish endpoint-decidable cases from poses on which both structural metrics are assigned. A known failure of one component can determine a combined failure even when another component is unassigned. Repeated-score variability uses the same three inference repeats. Matching cohort identities and endpoint definitions does not harmonize receptor preparation, candidate budgets, training exposure or the cofolding task; 341 of 558 FoldBench interfaces share a PDB accession with the docking-training inventory (Table S16). Evaluator and compound-library provenance accompany the source data.

The combined endpoint remains unresolved for FoldBench: 0/0/0; OpenBind: 0/0/0 complexes across the three repeats, respectively. A known failure of one component resolves a combined failure without imputing another component. Table S15 uses refined, chirality-filtered top-1 selection from the original 100-pose banks.

Native conjunctions are determined for every case in all four strata. Native RMSD alone is unassigned for one accession-shared interface in each repeat; its conjunction is determined by the known LDDT-PLI failure. Accession sharing does not establish identity of the chain-ligand complex, and nonsharing does not establish independence from all predecessor training exposure.

S7.4 Baseline protocols

The Astex/PoseBusters comparison includes EFF-Dock, five locally run supplied-pocket learned or hybrid methods, and two published classical baselines (Table S17) [30, 31, 32, 33, 34, 35, 36, 37, 23]. The locally evaluated methods, including EFF-Dock, are repeated three times and reported as a mean with sample standard deviation; the classical rows are literature aggregates. GOLD and Vina values are recomputed from the authors’ deposited [PoseBusters result table](#), restricted to the 85 Astex and 308 PoseBusters identities used here. They retain the deposited evaluation and non-minimized condition; their poses were not re-evaluated with PoseBusters 0.6.5. Missing targets, unmappable poses, and evaluator failures remain in the complete 85- or 308-complex denominator. Each method retains the documented local upstream generation, selection and refinement procedure, including the preparation compatibility steps below; the comparison is not compute matched.

In the baseline labels, MDN means mixture density network and EC denotes the native error-correction step; these identify the source scoring and post-processing procedures rather than additional EFF-Dock components.

Starting ligand geometry was method-specific rather than harmonized. SigmaDock used independent ETKDGV3/MMFF conformers after restoring unspecified SDF double-bond stereochemistry from the supplied three-dimensional coordinates. SurfDock used independent ETKDGV2 conformers,

Table S14: Benchmark-specific comparison with published models. PhiBench source rows are top-5 of 40 pocket-conditioned predictions; n is each method’s reported evaluable denominator. EFF-Dock rows use the local PhiBench-derived cohort and fixed-receptor RMSD, so equal sizes do not establish identical cases. FoldBench source rows are cofolding, while EFF-Dock uses a supplied holo receptor; their common structural conjunction requires binding-site-superposed RMSD $< 2 \text{ \AA}$ and LDDT-PLI > 0.8 . FoldBench $n = 558$ denotes the released group, not a verified source-model scoreable denominator. OpenBind uses the official 802-case follow-on subset, RMSD $\leq 2 \text{ \AA}$ and PoseBusters validity, with LDDT-PLI ≥ 0.8 for the additional conjunction. Source PhiBench RMSD requires PAL-RMSD $< 2 \text{ \AA}$; EFF-Dock uses symmetry-aware RMSD $< 2 \text{ \AA}$ and the 27-check conjunction. The 40-pose EFF-Dock prefix is selected from the stored 100-pose bank. Dashes denote unreported metrics. Rows retain their distinct tasks and preparation protocols.

PhiBench: pocket-conditioned source results and local redocking				
Method	n	Selection	RMSD (%)	RMSD-PB (%)
PhysDock	206	Top-5/40	83.0	77.7
SurfDock	184	Top-5/40	71.7	71.2
Interformer	180	Top-5/40	68.3	63.3
Uni-Mol Docking V2	184	Top-5/40	53.3	52.2
DiffDock-L	186	Top-5/40	39.8	35.5
Glide	185	Top-5/40	25.4	24.3
Vina	198	Top-5/40	23.7	22.7
DiffDock	186	Top-5/40	30.6	25.8
EFF-Dock (local)	206	Top-5/40 prefix	72.5 ± 5.0	70.4 ± 4.8
EFF-Dock (local)	206	Top-5/100	78.2 ± 0.5	76.1 ± 0.6
FoldBench: cofolding references and supplied-holo redocking				
Method	n	Selection	RMSD (%)	RMSD-LDDT (%)
AlphaFold 3	558	Native	–	64.9
Boltz-1	558	Native	–	55.0
Chai-1	558	Native	–	51.2
HelixFold 3	558	Native	–	51.8
Protenix	558	Native	–	50.7
OpenFold 3 (preview)	558	Native	–	44.5
EFF-Dock (holo)	558	Top-1/100	75.4 ± 0.5	71.4 ± 1.6
OpenBind: holo redocking, official follow-on subset				
Method	n	Selection	RMSD-PB (%)	+LDDT (%)
GNINA	802	Top-1	69.5	66.6
EFF-Dock	802	Top-1/100	56.4 ± 0.5	48.3 ± 0.8

Table S15: Top-1 benchmark-specific results and structural-score availability. Entries are percent mean $\pm$ sample SD over three fixed-weight inference repeats. The full-cohort rows count confirmed successes over every complex. Conditional rows retain only selected poses with both native RMSD and LDDT-PLI assigned; their per-repeat counts are listed in seed order. This availability-based subset is not a common case mask across models. RMSD denotes native RMSD success alone ($< 2 \text{ \AA}$ for FoldBench; $\leq 2 \text{ \AA}$ for OpenBind). RMSD-PB additionally requires PoseBusters validity. Combined requires RMSD and LDDT-PLI > 0.8 for FoldBench, and RMSD, PoseBusters validity and LDDT-PLI ≥ 0.8 for OpenBind, on the same selected pose. Dashes indicate an endpoint not reported for that benchmark.

Dataset	Denominator	n	RMSD	RMSD-PB	Combined
FoldBench	Full cohort	558	75.4 ± 0.5	–	71.4 ± 1.6
FoldBench	Both scores assigned	557,557,557	75.6 ± 0.5	–	71.5 ± 1.6
OpenBind	Full cohort	802	56.4 ± 0.5	56.4 ± 0.5	48.3 ± 0.8
OpenBind	Both scores assigned	802,802,802	56.4 ± 0.5	56.4 ± 0.5	48.3 ± 0.8

Table S16: FoldBench performance in fixed docking-training accession and release-date strata. All rows use the refined primary selector and the full stratum denominator. Values are percent mean $\pm$ sample SD across three inference repeats. Accession sharing is assessed against the 47,277-sample docking-training inventory. Fixed-receptor RMSD and RMSD–PoseBusters use the primary evaluation; native RMSD is binding-site-superposed and symmetry-aware, and the native conjunction additionally requires LDDT–PLI > 0.8 . The later-release slice uses a prespecified 30 June 2024 boundary and is included in the accession-unshared group. These outcome-blind strata describe provenance rather than novel pockets or a causal effect of training exposure.

Stratum	n	Fixed RMSD	RMSD–PB	Native RMSD	Native both
All interfaces	558	75.3 ± 0.5	73.8 ± 0.6	75.4 ± 0.5	71.4 ± 1.6
Training accession shared	341	76.6 ± 0.7	76.3 ± 0.7	76.6 ± 0.7	72.8 ± 1.9
Training accession unshared	217	73.1 ± 2.4	69.7 ± 2.6	73.6 ± 2.4	69.1 ± 2.1
Released after 30 June 2024	66	69.7 ± 1.5	68.2 ± 1.5	71.2 ± 1.5	68.7 ± 4.6

Table S17: Generation, selection, and receptor settings used in the Astex/PoseBusters comparison.

Arm	Generation	Selection and refinement	Receptor and pocket context	Evidence
EFF-Dock	100 candidates; 10 late-biased Euler intervals	Confidence ranking after chirality-mismatch removal; no in-ODE drift; energy-based post-sampling refinement	Fixed holo receptor; reference-derived point center from residue nodes within 8 Å of the ligand; 10 Å docking crop	Three repeats
SigmaDock	Euler $S = 25$; 40 independent samples per run	Official Vinardo plus seven-check PoseBusters selector; no coordinate minimization	Fixed holo receptor; official 5 Å ligand-proximity residue graph with official pocket/noise defaults	Three repeats
DiffDock-Pocket	30 diffusion steps; 40 poses	Released confidence ranker; no optional relaxation	Holo-aligned predicted receptor; cognate supplied pocket with the native ligand-contact $C\alpha$ centroid/crop procedure	Three repeats
RLDiff RL++	20 steps; 40 poses	smina minimization followed by GNINA ranking	Fixed holo receptor; cognate pocket obtained from a 5 Å ligand-contact $C\alpha$ set and a ligand-size-dependent crop	Three repeats
DiffBindFR + MDN/EC	40 poses; native 20 effective diffusion steps; batch size 16	Official smina error correction followed by MDN ranking	Holo receptor input; native pocket side-chain torsions on a fixed backbone; cognate-ligand coordinates define the upstream 12 Å MDN scoring graph	Seeds 0, 1, 2
SurfDock	20 diffusion steps; 40 poses	Native force-field optimization with released failure fallback, followed by MDN confidence ranking	Fixed supplied-pocket receptor context under the released local-run protocol	Three repeats
GOLD	Published aggregate; no local generation	Reported Top-1 value	Supplied-pocket literature baseline; per-complex site construction unavailable in the aggregate result	Published aggregate
AutoDock Vina	Published aggregate; no local generation	Reported Top-1 value	Supplied-pocket literature baseline; per-complex docking box unavailable in the aggregate result	Published aggregate

with MMFF optimization in its embedding-failure fallback. Retained DiffDock-Pocket, RLDiff and DiffBindFR paths used supplied local ligand geometry before their native pose randomization. For DiffDock-Pocket and RLDiff, this description is supported by retained source and runner configuration; complete resolved-argument records for every execution were not recovered. These preparation differences limit attribution of performance differences to model architecture alone.

DiffBindFR’s retained generation path samples pocket side-chain torsions while keeping the backbone fixed. The common PoseBusters evaluation instead uses the frozen holo receptor for its selected ligand, as for the other local methods. This endpoint therefore differs from evaluating the jointly sampled protein–ligand structure and may penalize clashes relieved by native side-chain motion.

SurfDock attempts force-field optimization before evaluating its native confidence score. Failed candidates undergo the upstream pocket-receptor fallback and are retained if minimization remains unsuccessful; minimization exceptions also retain candidates for scoring. Thus enabling force optimization does not imply successful minimization of every candidate. The common evaluator computes symmetry-aware full-heavy-atom RMSD and official PoseBusters pass-all for every locally evaluated learned or hybrid method, retaining missing targets and unmappable outputs as denominator failures. The main-text comparison displays same-pose RMSD–PoseBusters success as the solid portion of each bar and RMSD-only success as the total height; the classical rows retain their published evaluation procedures.

S7.5 Atom mapping and failure handling

Primary RMSD eligibility requires a complete heavy-atom bijection between the model-input ligand and the reference ligand: atom counts must agree, every heavy atom must be mapped once, and mapped elements and constitutional connectivity must agree. Matching attempts proceed through strict, charge-neutralized, and five-second element-constrained maximum-common-substructure routes in that order. For an equal-topology pair, RDKit `CalcRMS` evaluates symmetry-equivalent atom permutations without first aligning the predicted pose to the reference, and the accepted map must still cover the complete heavy-atom entity. Partial maximum-common-substructure mappings are used only as diagnostics and do not enter the primary endpoint.

Physical validity is evaluated with PoseBusters 0.6.5 using the `redock` configuration [23]. Astex and PoseBusters use the native PoseBusters 0.6.5 `redock` pathway. PhiBench-derived, FoldBench and OpenBind use a compatibility adapter for the molecule constructed from InChI as the internal-energy reference. A mapped SMILES round trip restores its representation, with checks on atom count, atom identities, connectivity, canonical isomeric SMILES and InChI. Predicted and experimental pose coordinates, acceptance thresholds and the 27 check definitions are unchanged. This cohort-specific representation normalization is part of the reported evaluation protocol. Pass-all is the logical conjunction of all 27 non-RMSD checks on the same selected pose. Failed or unavailable reported check results make pass-all false. A non-passing result therefore does not, by itself, establish a specific geometric defect. Unsupported chemistry, preprocessing failure, incomplete atom mapping, sampling failure, non-finite coordinates, missing output, evaluator failure, and resource failure remain in the cohort denominator and are reported by reason.

For each complex, receptor and ligand identities, input preparation, and pocket provenance are verified before docking. Reference-ligand coordinates are removed from model and guidance inputs, and one independently generated conformer is prepared. Raw candidates are saved before refinement; both raw and refined banks retain original candidate indices and are scored without RMSD or validity labels. The confidence-selected pose is then used for complete heavy-atom RMSD mapping and PoseBusters `redock` evaluation. PhiBench-derived retains the three recovered systems, OpenBind retains all 925 annotated rows, and FoldBench retains its recorded International Chemical Identifier (InChI)-compatibility outcomes in the corresponding denominators [38].

S7.6 Metrics and uncertainty

Confidence discrimination uses negative predicted RMSD as the score and $\text{RMSD} < 2 \text{ \AA}$ as the binary label within each candidate bank. AUROC and average precision are macro-averaged over banks containing both positive and negative candidates; all-success and all-failure banks are excluded from these two metrics. Spearman correlation uses average ranks for ties and excludes banks with a constant score or constant RMSD. The numerical source tables report the contributing bank counts for

each repeat. Raw and refined discrimination summaries can therefore contain different sets of contributing banks. Reliability curves use the saved predictions without fitting a calibration transform; the success-probability head is separate from the predicted-RMSD score used for selection. RMSD means root-mean-square deviation: the square root of the mean squared heavy-atom position error over a complete atom correspondence. Primary scoring minimizes over admitted symmetry-equivalent correspondences without independent ligand alignment. AUROC is area under the receiver-operating-characteristic curve and measures discrimination over score thresholds; average precision summarizes precision weighted by recall increments. Spearman correlation, labelled ρ in the figure, compares candidate ranks; this statistical ρ is distinct from the steric radius scale. Macro-averaging weights each contributing bank equally. The 2 Å endpoint follows the cited docking benchmarks, and the evaluation configuration defines the 27-check PoseBusters conjunction. Lower RMSD and regret are preferable; larger success, correlation, AUROC and average precision are preferable. LDDT-PLI is the local distance difference test for protein–ligand interactions; BiSyRMSD is binding-site-superposed, symmetry-aware ligand RMSD.

The primary endpoint is top-1 redocking success: symmetry-aware heavy-atom RMSD below 2 Å, measured without alignment. Secondary endpoints are top-5 and oracle-100 coverage, selected mean and median RMSD, official PoseBusters 27-check non-RMSD pass-all, and RMSD–PoseBusters success for the same pose. For confidence-ranked top- k coverage, a complex succeeds if any of the first k ranked candidates has RMSD below the threshold. Oracle-100 evaluates the complete bank regardless of ranking. For generation-prefix analyses, only the first N saved poses are retained, after which top-1 is reselected within that prefix; this reuses an existing 100-pose bank and is not a separately executed N -pose campaign. Raw and refined outcomes and selection-condition transitions are paired by complex ID and summarized by gain/loss counts. Continuous RMSD summaries use complete cases and are reported with their coverage.

Ligand-binding chains in the training-relatedness analysis are defined by protein–ligand heavy-atom contacts within 6 Å. Maximum sequence identity is query-normalized: identical known residues under one-to-one chain assignments are divided by the total observed query binding-chain length, including unaligned residues and unmatched query chains. Exact ligand identity uses canonical heavy-atom isomeric SMILES with explicit-hydrogen normalization; non-identical ligands are compared by radius-2, 2,048-bit Morgan-fingerprint Tanimoto similarity without chirality [39, 1]. Fingerprint radius 2 uses atom environments out to two bonds, and 2,048 is the hashed bit-vector length. Tanimoto is the number of shared on-bits divided by the number in their union, ranging from zero to one; identical fingerprints do not alone establish exact chemical identity. All relatedness panels use the same 47,277-member docking-training reference; each metric uses its available, parseable ligand or chain records. Unparseable ligands are not certified as negative matches. The figures describe overlap with this docking-training set. An additional comparison includes the 25 confidence-training identities absent from it, giving a combined training set of 47,302 samples (Table S18). Neither analysis reconstructs the complete ancestral-checkpoint lineage or establishes causal memorization.

Adding the 25 confidence-only samples left external exact ligand matches, Morgan Tanimoto maxima and stringent-subset membership unchanged (Table S18). Conservative sequence bounds preserved external maximum identities and 1,075 of 1,076 validation maxima. The remaining validation identity was bounded at 13.51–14.41%, below the 30% threshold. Bounds based on reference-chain lengths or standard-residue longest common subsequences were maximized over one-to-one chain assignments with the full query-length denominator. All annotated reference chains enter the bounds as a superset of contact-qualified chains. The reported identity metric counts unknown residues as non-identities; allowing unknown residues in the added samples to match any query residue also preserved stringent membership. These bounds do not resolve unknown residues, sequence-version differences or ancestral training exposure.

Unless stated otherwise, uncertainty is the sample standard deviation across three inference repeats on the same complex set, with identical trained weights. Astex and PoseBusters use a fixed seed-zero generation conformer; in PhiBench-derived, FoldBench and OpenBind, conformer generation follows the per-complex sampling seed. Repeat variability in the latter cohorts therefore includes starting-conformer variation. None of these repeats represents independent training or target-population resampling. For the paired improvement analysis, per-complex differences are first averaged across repeats and then resampled with 2,000 percentile-bootstrap draws; the complex bootstrap resamples complexes, whereas the PDB-cluster bootstrap resamples exact-PDB groups and reports a complex-

Table S18: Effect of adding the 25 confidence-only training samples to the docking-training reference. Sequence counts are certified with conservative bounds over the frozen sequence records. Stringent membership uses no observed exact ligand match, maximum Morgan Tanimoto below 0.5 and sequence identity below 30%; all listed memberships are unchanged.

Dataset	n	New exact ligand matches	Tanimoto maxima increased	Sequence maxima unchanged	Stringent subset
Astex Diverse Set	85	0	0	85	2
PoseBusters v2	308	0	0	308	4
PhiBench-derived	206	0	0	206	7
FoldBench	558	0	0	558	9
OpenBind	925	0	0	925	0
Docking validation	1,076	0	2	1,075	448

weighted mean [40]. The paired baseline-difference analysis likewise uses 2,000 percentile draws but resamples exact PDB-accession groups; every retained Astex and PoseBusters complex has a distinct PDB accession, so group and complex resampling coincide for those two cohorts. Percent rates are successful complexes divided by the stated denominator, multiplied by 100; a percentage-point difference subtracts two percent rates without dividing by the baseline. Sample SD uses the three-repeat mean and divisor $3 - 1$; it is distinct from a bootstrap confidence interval. The 95% percentile interval uses the 2.5th and 97.5th percentiles of 2,000 resampled estimates. The draw count specifies Monte Carlo resampling of existing outcomes. Cohort sizes, exclusions and success counts are observed inventory or outcome counts; versions, dates and PDB/CCD identifiers specify provenance rather than numerical hyperparameters.

S7.7 Ligand complexity and candidate-bank size

Performance stratified by ligand size, rotatable-bond count, and rigid-fragment count is shown in Fig. S1. The cumulative-ranking and generation-prefix analyses in Figs. S2 and S3 distinguish coverage gained by adding candidates from performance gained by ranking them correctly.

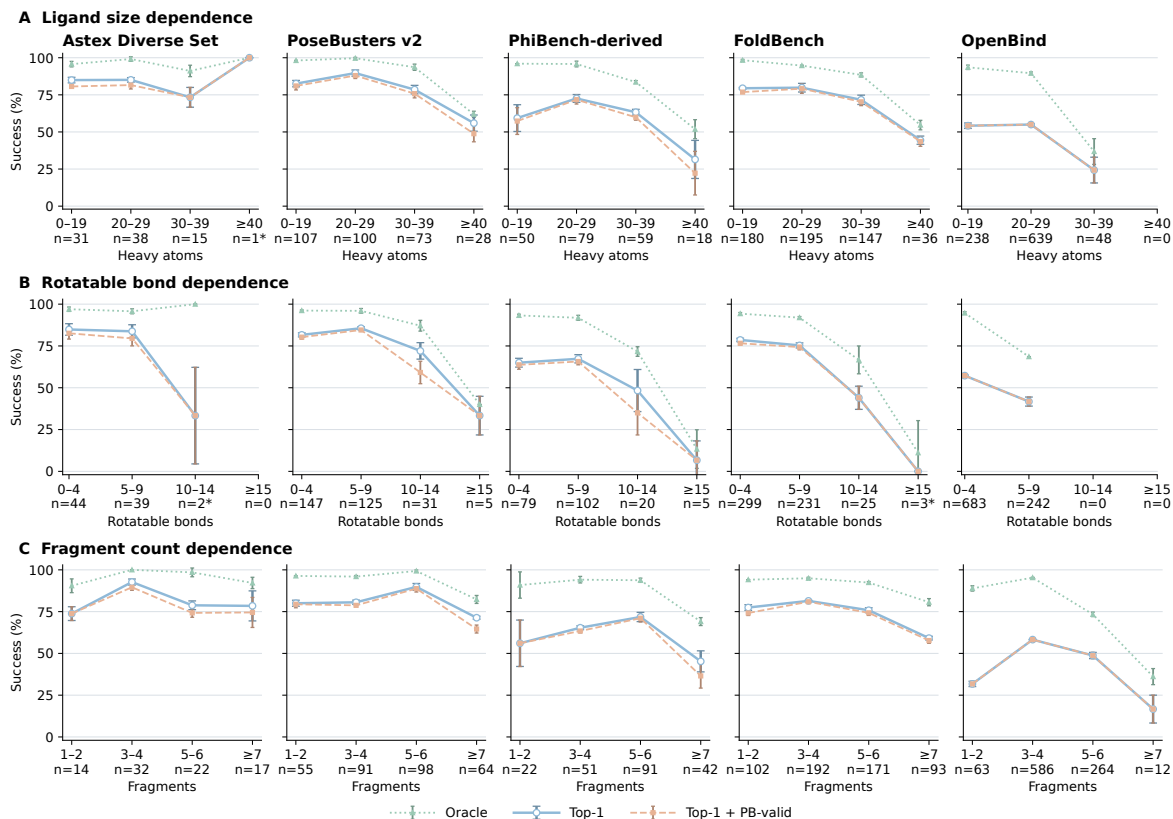


Figure S1: **Docking performance across ligand-complexity strata.** EFF-Dock results use refined candidates and chirality-filtered selection. The five benchmarks are grouped by (A) ligand heavy-atom count, (B) strict rotatable-bond count, and (C) saved rigid-fragment count. Top-1 denotes RMSD $< 2 \text{ \AA}$; Top-1 + PB-valid requires the same selected pose to pass PoseBusters; Oracle searches all 100 refined candidates using RMSD only. Points show mean $\pm$ sample SD across three repeats. Labels give unique-complex counts, and asterisks mark bins with fewer than five complexes.

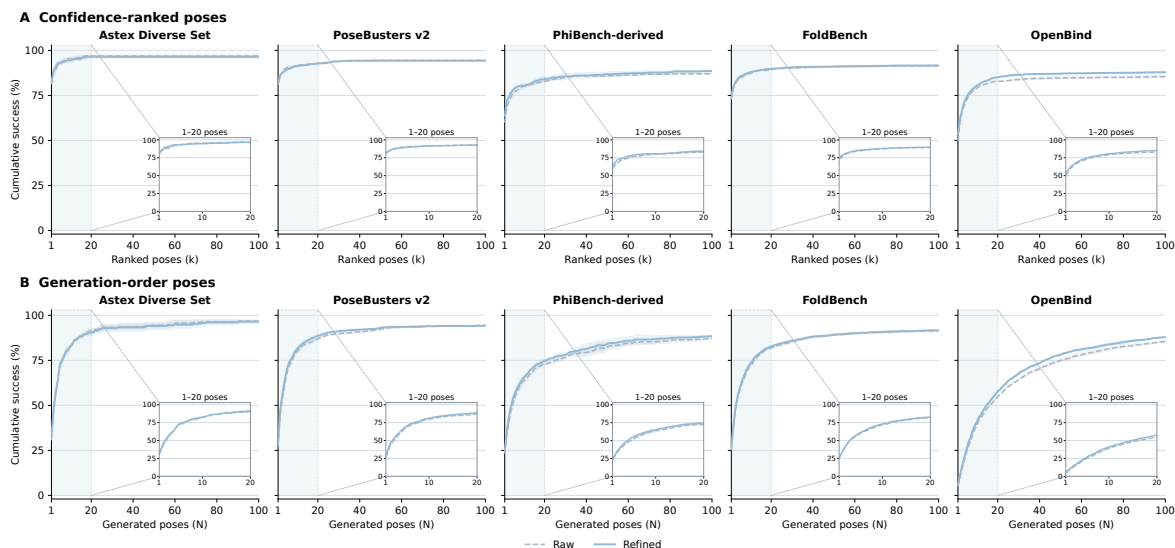


Figure S2: **Cumulative pose success under confidence ranking and generation order.** EFF-Dock results are shown for (A) the first k candidates after unfiltered confidence ranking of the complete bank and (B) the first N candidates in saved generation order without confidence sorting. Solid and dashed lines denote refined and raw banks; bands show ± 1 sample SD across three repeats. Insets expand candidates 1–20. Both orderings reach the full-bank RMSD-only oracle at 100 poses; PoseBusters validity is not included.

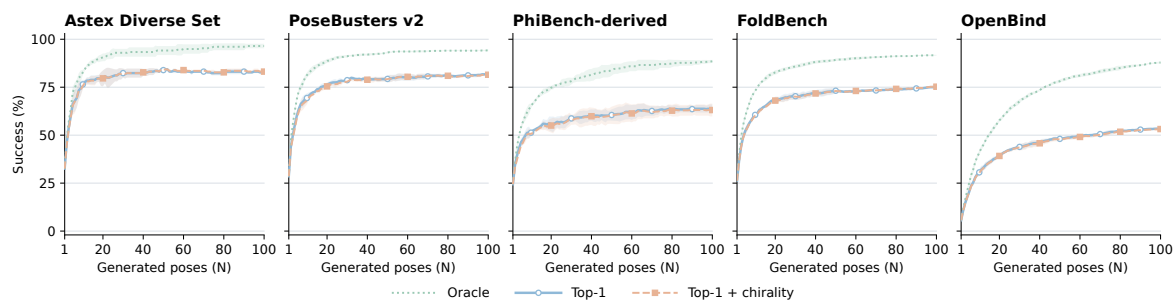


Figure S3: **Selected-pose success as a function of available candidate count.** The first N refined candidates are retained from each EFF-Dock 100-pose bank. Top-1 reselects the minimum predicted-RMSD pose, Top-1 + chirality applies the input-chirality mask with the production fallback, and Oracle tests whether the prefix contains any RMSD-successful pose. Lines and bands show mean ± 1 sample SD across three repeats. The curves reuse saved prefixes and are not independently executed smaller campaigns or runtime measurements.

S7.8 Resource measurements

The docking model has 5,719,970 parameters; the confidence model has 9,403,274. These counts sum parameter-tensor elements in the respective networks. Runtime uses wall-clock seconds per complex, and plotted GiB denotes 2^{30} bytes of CUDA allocated memory; GPU model numerals are device names.

Runtime and memory measurements are shown in Figs. S4 and S5. Profiling used mixed RTX A5000, RTX 6000 Ada and RTX PRO 6000 devices. Recorded times include the original refinement and confidence scoring but exclude subsequent refinement and rescoring passes; they do not estimate the end-to-end cost of the primary protocol or support a controlled hardware comparison.

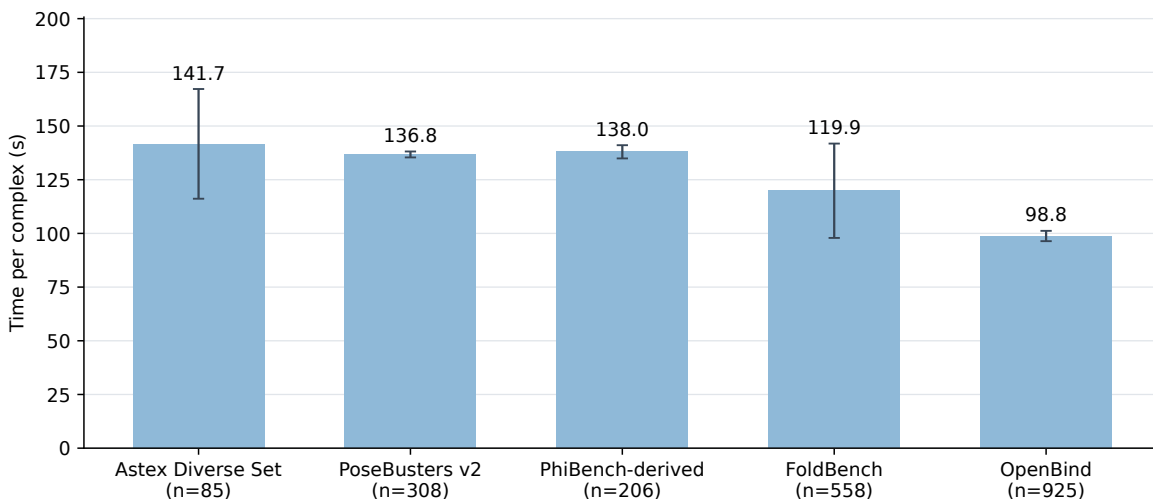


Figure S4: **Initial candidate-acquisition pipeline runtime.** Mean wall time per complex for the unguided 100-pose, 10-step evaluation. Runtime covers setup, generation, original refinement and confidence scoring, and I/O. It excludes official PoseBusters evaluation and subsequent refinement and rescoring passes, so it is not the end-to-end runtime of the primary results. Whiskers show sample SD across three fixed-weight inference repeats. Mixed devices and cohort-specific process profiles preclude a controlled speed comparison.

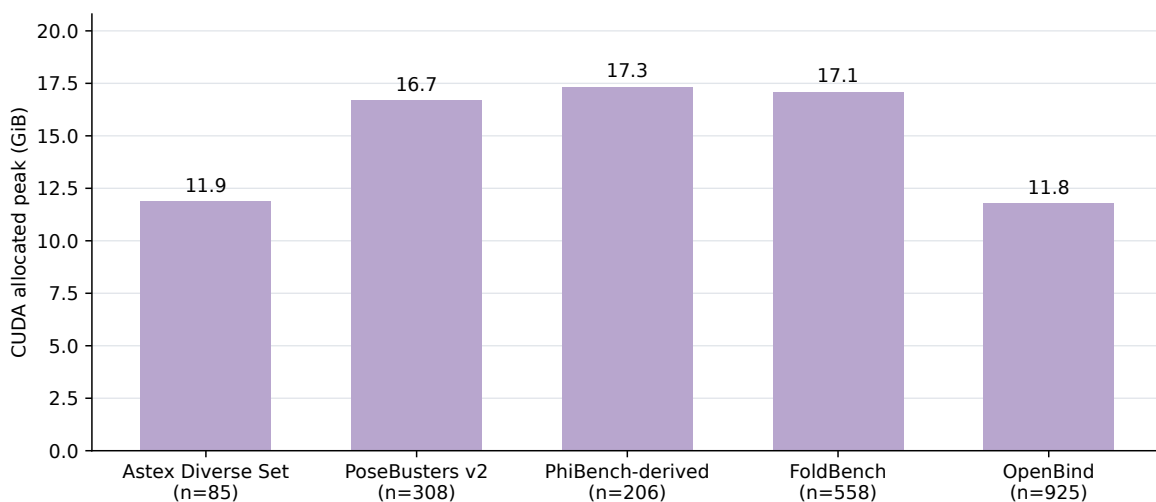


Figure S5: **Initial pose-generation process memory.** Maximum CUDA allocated-memory peak across three repeats of the unguided 100-pose, 10-step evaluation. This is the original generation/evaluation process allocator peak, not reserved memory or a whole-pipeline peak. Astex and PoseBusters used candidate-only generation processes, whereas PhiBench-derived, FoldBench and OpenBind processes included initial confidence scoring. Separate post-generation refinement and rescoring processes are excluded. Mixed devices and these different process profiles preclude a controlled comparison of sampling-only memory.

S7.9 Training-set relatedness

Training-set relatedness varied across cohorts (Figs. S6–S8). All OpenBind complexes occupied the highest sequence-identity bin. FoldBench had the greatest same-training-sample overlap: 63.4% shared an exact ligand match and at least 70% sequence identity, whereas this conjunction was absent in docking validation. PDB-accession sharing is distinct from joint ligand–sequence overlap because an accession can contain different chains or ligands; accession correspondence was unavailable for OpenBind. Neither ligand nor sequence relatedness showed a consistent monotonic association with success. In the overlap legend, same-sample means an observed exact ligand match and binding-chain identity of at least 70% in the same training sample; separate matches satisfy the two criteria in different samples without such a joint match. Ligand-only and sequence-only satisfy one criterion, and neither observed indicates absence of both observed matches. The 70% identity threshold is a descriptive overlap rule, distinct from PLINDER pocket-community membership. In the ligand-similarity bins, T denotes Tanimoto similarity, not the selected-success rate T used in failure decomposition.

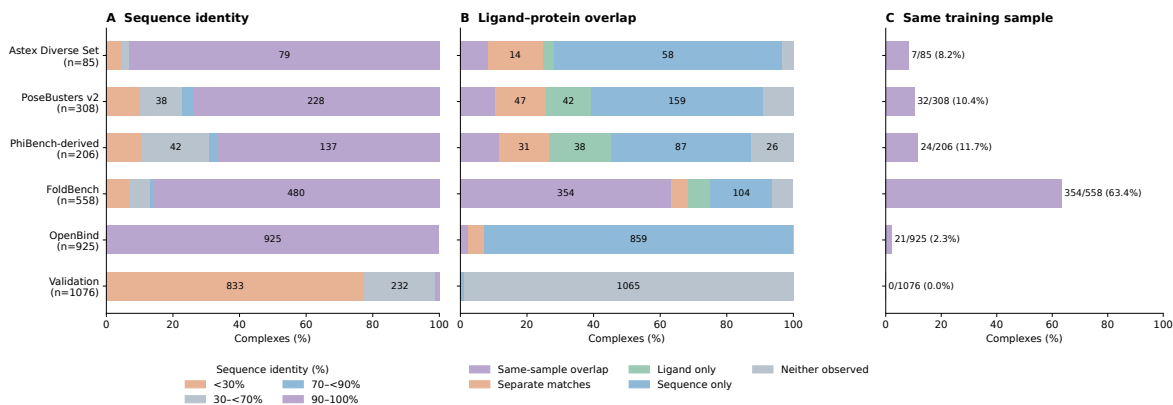


Figure S6: **Sequence and ligand relatedness to the docking-training set.** The five external cohorts and 1,076-complex docking-validation set are compared with 47,277 executed docking-training samples. (A) Maximum query-normalized binding-chain sequence identity. (B) Joint ligand and sequence overlap classified by whether the matches occur in the same training sample. (C) The same-sample overlap fraction with counts. Ligand-binding chains use protein–ligand heavy-atom contacts within 6 Å.

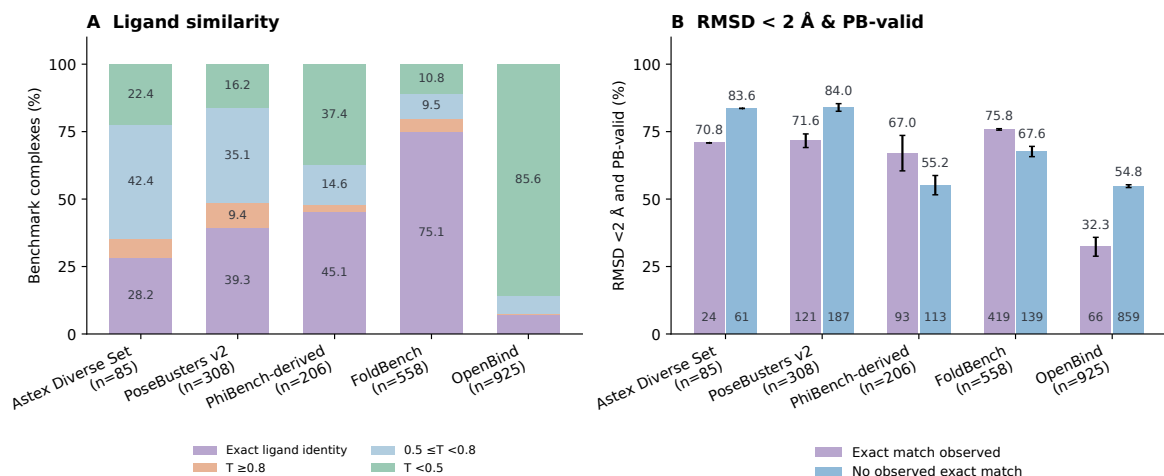


Figure S7: **Training-ligand relatedness and stratified docking performance.** (A) Exact heavy-isomeric ligand identity and nearest training-ligand Morgan-fingerprint Tanimoto similarity relative to the 47,277 executed docking-training samples. (B) EFF-Dock RMSD–PoseBusters success with and without an observed exact ligand match. Bars show mean $\pm$ sample SD across three repeats; group differences are descriptive and confounded by cohort composition.

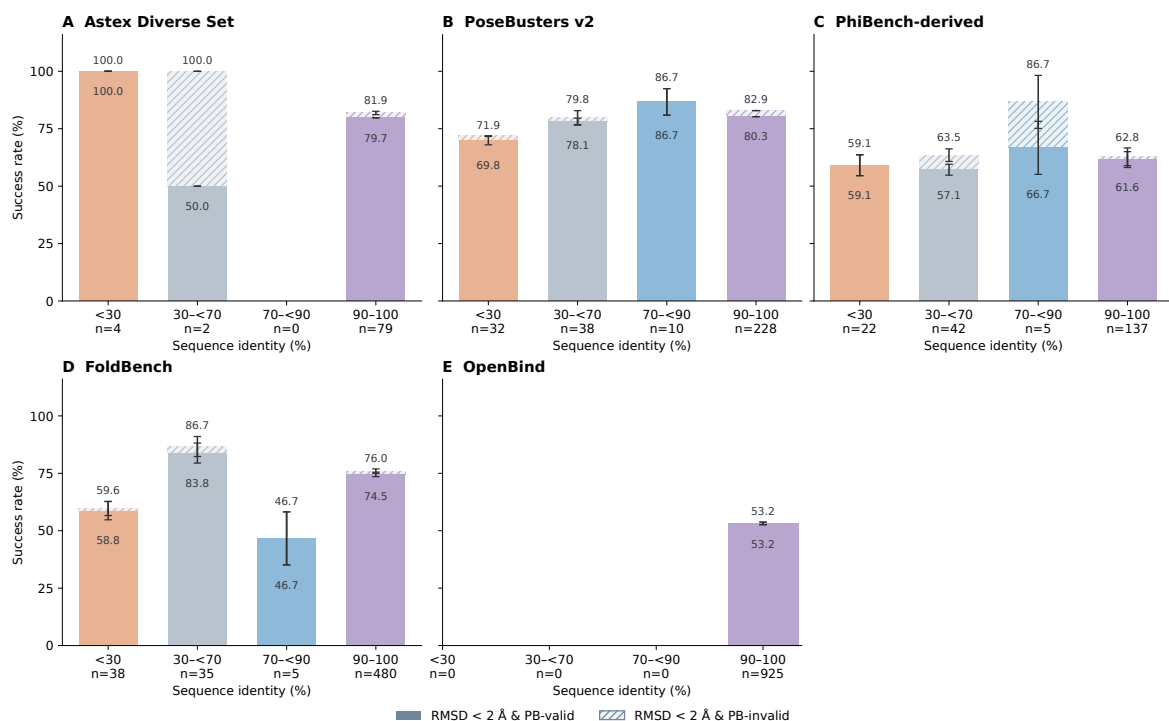


Figure S8: **Docking performance by training-set sequence identity.** Strata use maximum query-normalized binding-chain identity to the docking-training set. Solid bars show RMSD–PoseBusters success; hatching extends to RMSD-only success. Bars and whiskers show three-repeat means and sample SD; labels give stratum sizes, including empty and sparse groups.

S7.10 Paired improvements and failure decomposition

Figs. S9–S11 separate the paired effect of post-processing, the density of useful candidates, and the stage at which each complex fails.

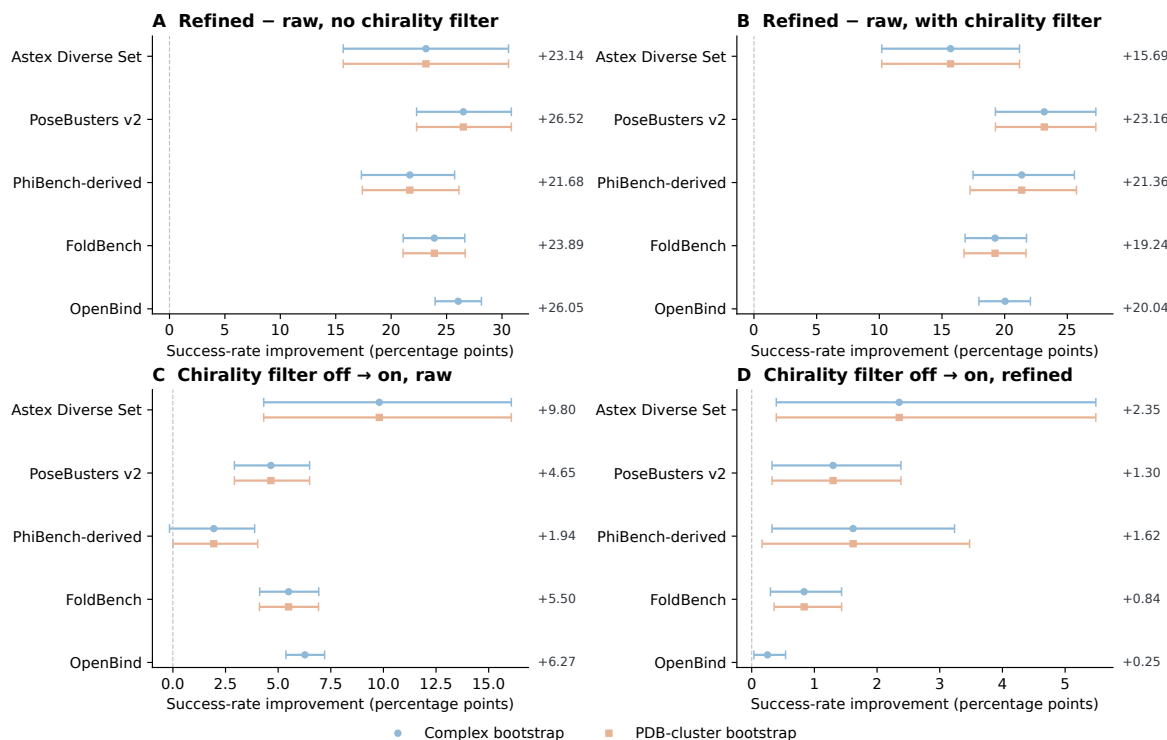


Figure S9: **Paired changes in docking success with refinement and chirality filtering.** EFF-Dock results compare refined versus raw candidates with and without the chirality filter and filter off versus on at each stage. The endpoint is RMSD < 2 Å and PoseBusters success. Per-complex paired differences are averaged across seeds before 2,000-draw percentile bootstrapping. Whiskers show complex and exact-PDB-cluster 95% intervals; OpenBind lacks a cluster interval.

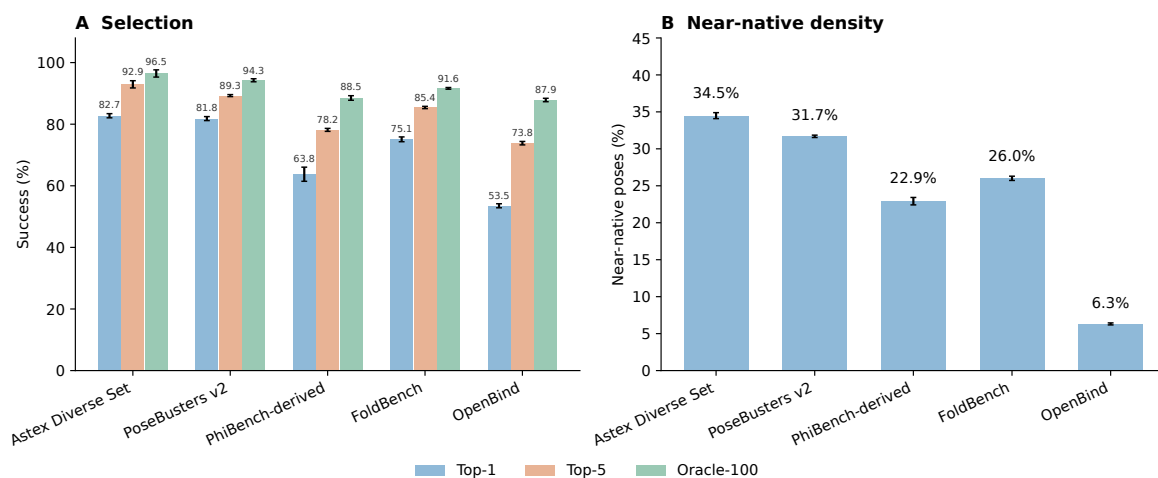


Figure S10: **Confidence selection and near-native candidate density.** (A) Top-1, top-5, and oracle-100 RMSD success for refined unguided banks. (B) Fraction of individual refined candidates with RMSD < 2 Å. Bars show mean ± sample SD across three repeats. Panel A is measured per complex, whereas panel B is measured per candidate.

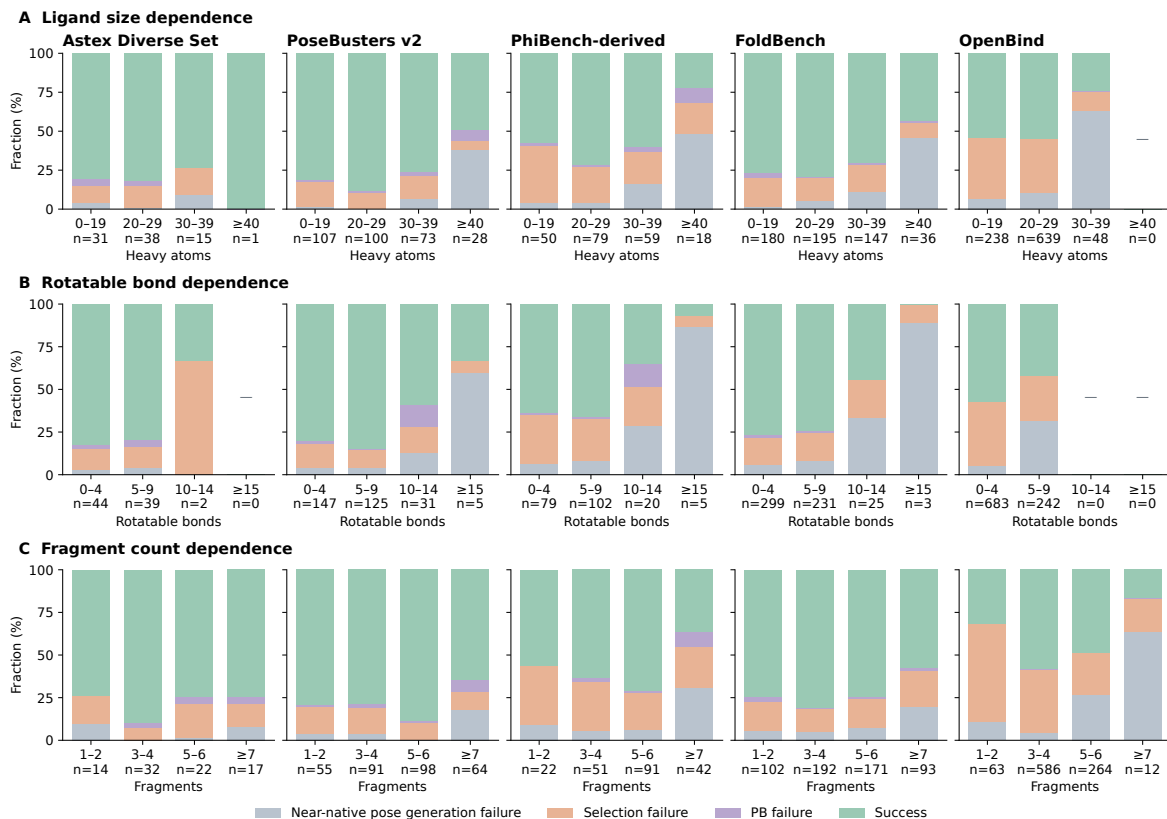


Figure S11: **Descriptive failure decomposition across ligand-complexity strata.** EFF-Dock, refined, chirality-filtered outcomes are partitioned into no RMSD-successful candidate, an available successful candidate not selected, selected RMSD success with PoseBusters failure, or RMSD–PoseBusters success. The fractions are the algebraic decomposition $100 - O$, $O - T$, $T - J$, and J of oracle coverage O , selected RMSD success T , and RMSD–PoseBusters success J . Generation failure refers to the refined bank and does not isolate the sampler.

S7.11 Confidence reliability and restricted evaluations

Fig. S12 complements the within-bank rank diagnostics in the main text with empirical reliability curves. The stringent slices in Fig. S13 are small, predefined evaluation subsets defined by the simultaneous ligand and sequence criteria and are therefore descriptive rather than population-level estimates. Probability bins have boundaries 0.1, 0.2, ..., 0.9; predicted-RMSD bins have boundaries 1, 2, 3, 4, 6 and 10 Å. Bin summaries pool candidates across repeats, with boundary values assigned to the higher bin. Near-native-count groups are 0, 1–5, 6–20, 21–50 and 51–100. Bins are fixed display strata. Identity < 30% and Tanimoto < 0.5, together with no observed exact ligand match, define the low-relatedness slice.

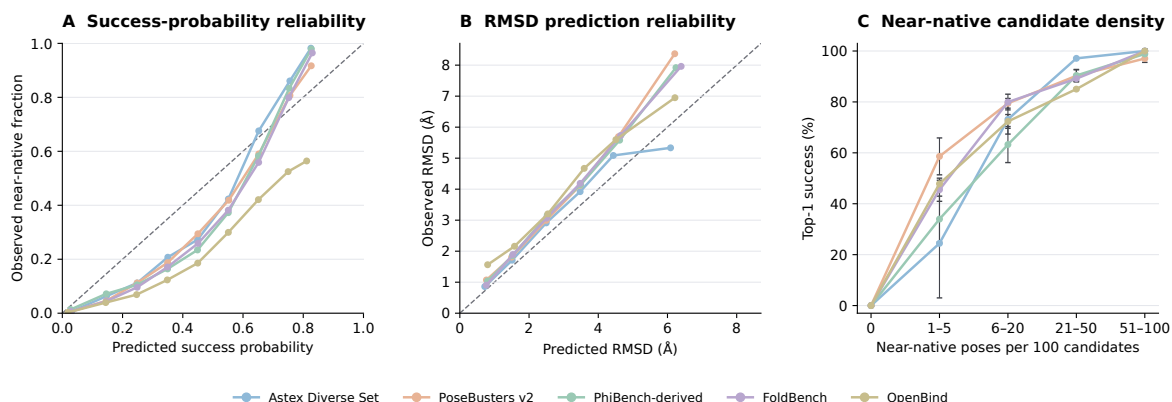


Figure S12: **Reliability of saved confidence predictions and candidate-density dependence.** EFF-Dock refined candidates are pooled across three repeats. (A) Empirical near-native fraction versus the saved success-head probability. (B) Mean observed versus predicted RMSD in fixed predicted-RMSD bins. No recalibration is fitted. (C) Primary chirality-filtered top-1 success versus the number of near-native candidates in the 100-pose bank. Candidate observations within complexes are correlated, and the production selector uses predicted RMSD rather than the auxiliary probability head.

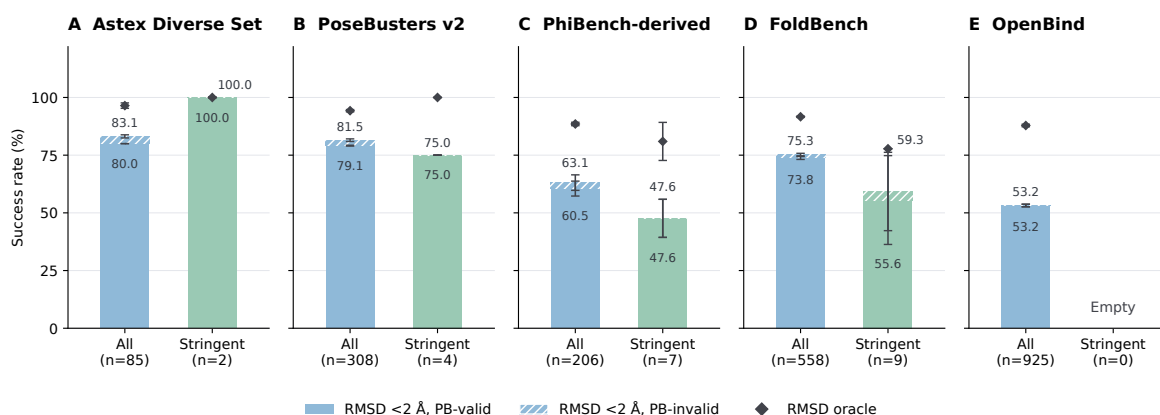


Figure S13: **Performance under simultaneous sequence and ligand restrictions.** The fixed subset requires maximum training-relative binding-chain identity < 30%, Morgan Tanimoto < 0.5, and no observed exact training-ligand match. Selected RMSD and RMSD–PoseBusters success and full-bank RMSD oracle are shown for EFF-Dock. Remaining counts are 2, 4, 7, 9 and 0 for Astex, PoseBusters, PhiBench-derived, FoldBench and OpenBind, respectively; the slices are descriptive.

S7.12 Paired baseline uncertainty and trajectory visualization

Fig. [S14](#) reports paired uncertainty for the locally executed baseline comparisons, and Fig. [S15](#) shows the saved one-pose trajectory used for the workflow illustration.

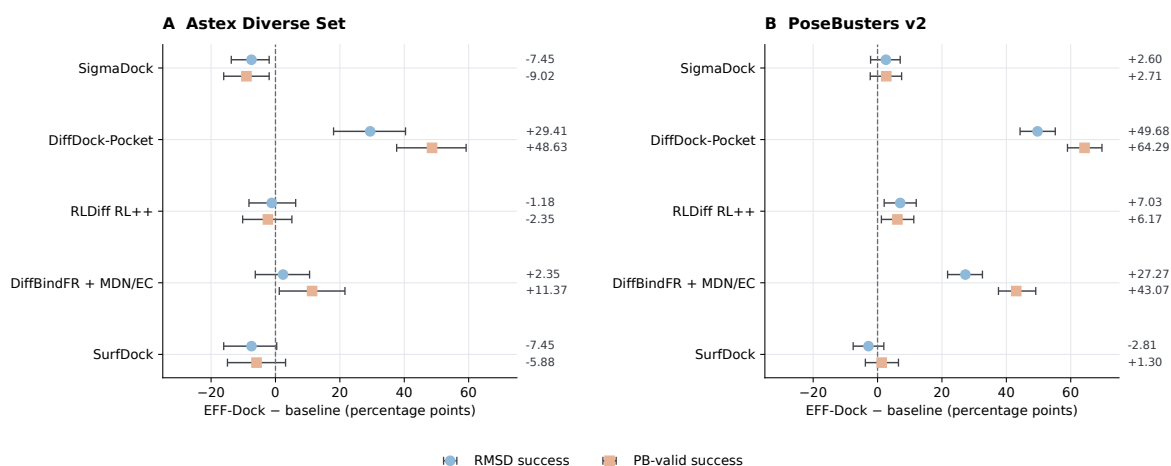


Figure S14: **Paired uncertainty relative to locally executed baselines.** EFF-Dock minus baseline percentage-point differences are shown for RMSD success and same-selected-pose RMSD–PoseBusters success on (A) Astex and (B) PoseBusters. Intervals are 2,000-draw percentile 95% paired bootstrap intervals over exact PDB-accession groups; every retained complex has a distinct accession in these cohorts. Literature-only rows are excluded. Positive values favor EFF-Dock, but differing budgets and native selectors prevent an equal-compute superiority claim.

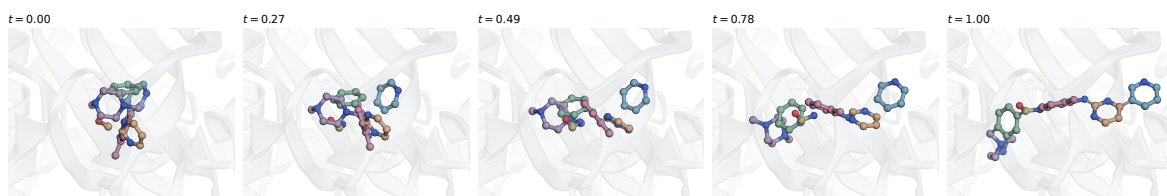


Figure S15: **Fragment motion along a saved ODE trajectory.** Astex 1T46–STI at five saved flow times in a fixed receptor frame. Colors track six rigid fragments; omission of early interfragment bonds is a display convention. Dimensionless flow time describes generation, separate from refinement. This prespecified seed-42, unguided one-pose illustration is separate from the 100-pose benchmark banks.

S7.13 Physical validity and representative structures

Physical-validity failures are reported in Fig. S16. Raw and refined selection may identify different candidates; matched-index transitions are limited to the selection-biased subset evaluated at both stages.

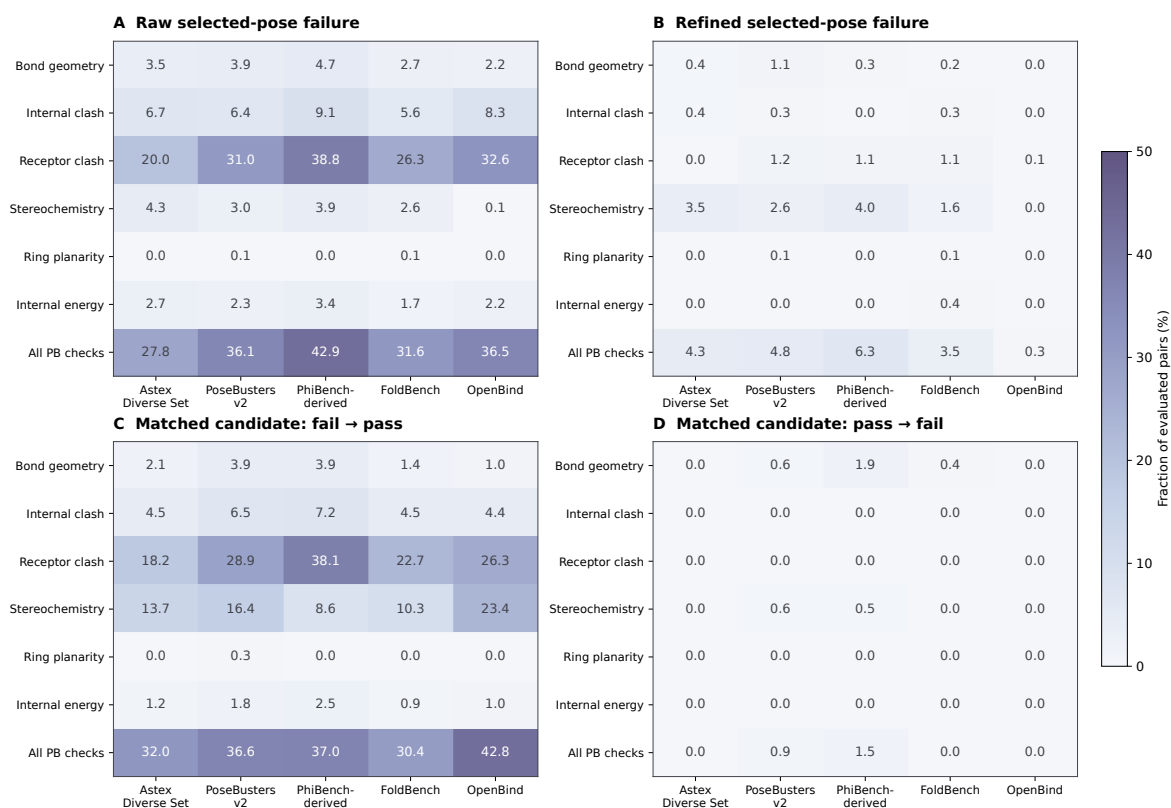


Figure S16: **Physical-validity failures before and after refinement.** (A,B) Heatmaps report failure percentages among the primary selected raw and refined poses; the selected candidate may differ between stages. (C,D) Fail-to-pass and pass-to-fail percentages use candidate indices with saved official PoseBusters evaluations at both stages. Rows summarize bond geometry, internal clash, receptor clash, stereochemistry, ring planarity, internal energy, and all 27 non-RMSD checks. Categories overlap, and the matched subset is selection biased.

Fig. S17 illustrates selected-pose success, refinement rescue and selection failure in each cohort. These cases are selected by a deterministic rule rather than random sampling. Each rescue shows the largest same-candidate RMSD decrease satisfying raw $\text{RMSD} \geq 2 \text{ \AA}$, refined $\text{RMSD} < 2 \text{ \AA}$, and selected-pose PoseBusters validity; it should not be interpreted as a typical effect.

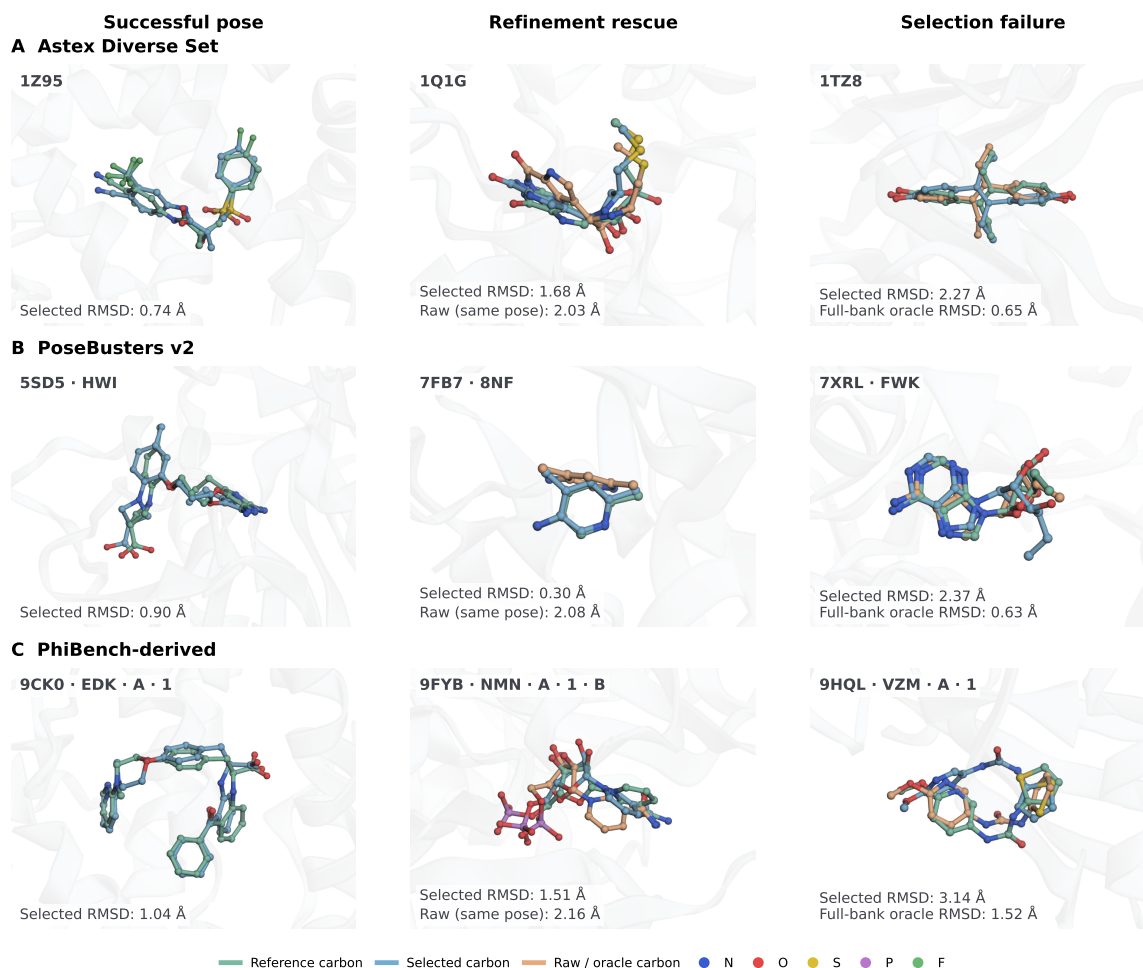
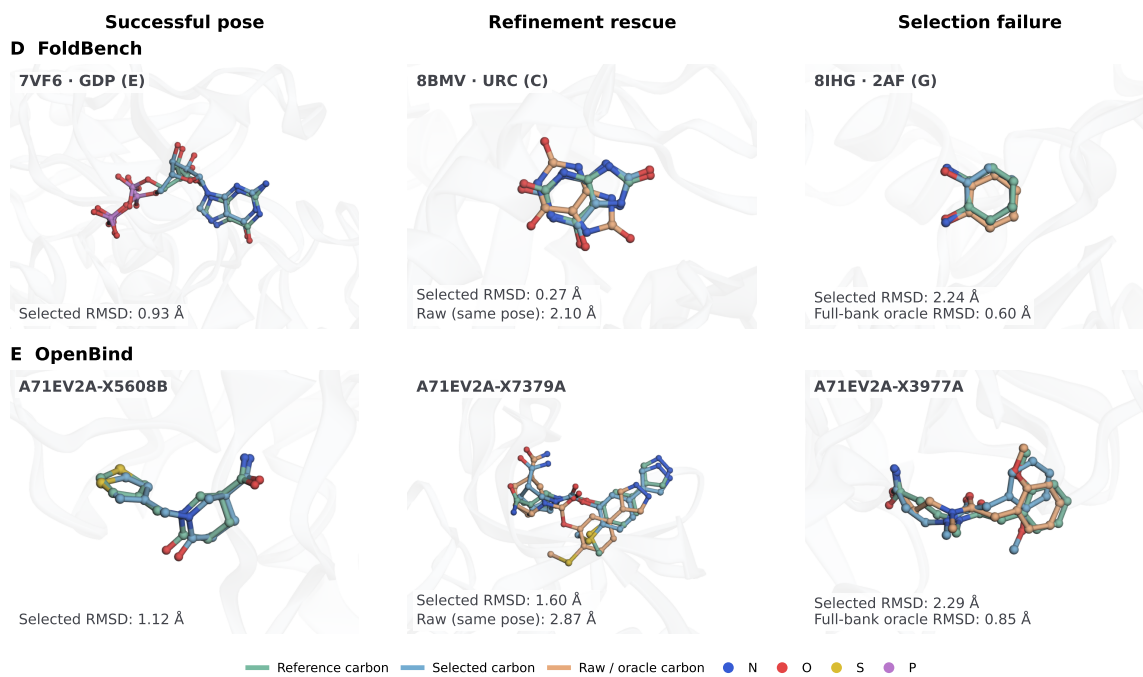


Figure S17: **Dataset-specific examples of success, refinement rescue, and selection failure.** (A–C) Astex Diverse Set, PoseBusters v2 and PhiBench-derived; (D,E) FoldBench and OpenBind appear on the continuation page. Rescue panels show the largest same-candidate RMSD decrease meeting the stated success and validity criteria in each dataset. Successful and selection-failure cases are deterministic lower-median examples from the remaining qualifying records. Rescue panels therefore show extremes rather than typical effects. Coordinates remain in the receptor frame without independent ligand alignment. A failed primary selection is compared with the minimum-RMSD pose from the full saved bank; this comparator is not required to pass the chirality screen or PoseBusters and does not by itself isolate a confidence-ranking failure.



Supplementary Fig. S17 (continued). (D,E) FoldBench and OpenBind examples, with the same structure, color and candidate-selection conventions as panels A–C.

S7.14 Eligibility-aware failure decomposition

Fig. S18 partitions outcomes using the same effective eligible set as the primary selector, including its unfiltered fallback. Candidate-bank success is defined solely by RMSD. Official PoseBusters labels were not collected for every candidate, so neither the full-bank nor the eligible-bank oracle establishes attainable RMSD–PoseBusters success.

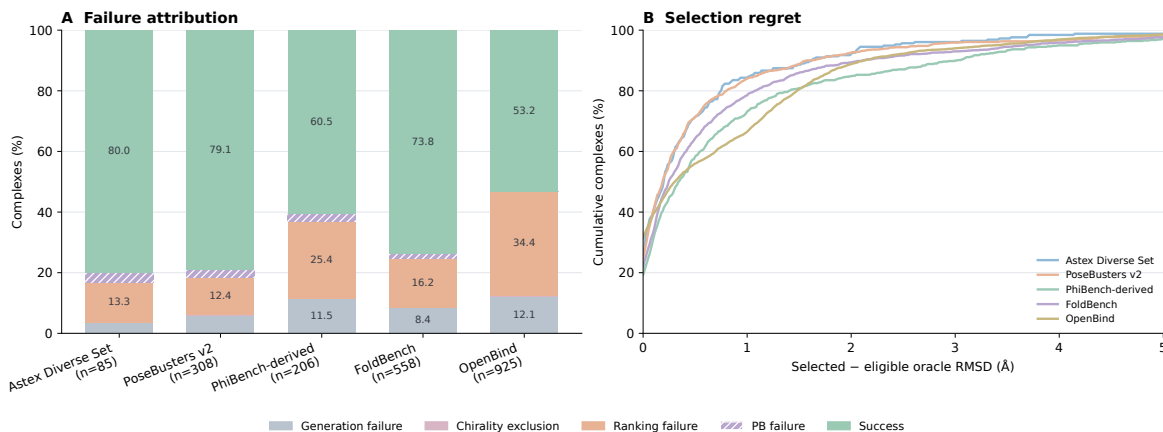


Figure S18: **Eligibility-aware selection bottlenecks and regret.** EFF-Dock results use refined 100-pose banks and the frozen chirality-filtered confidence selector. (A) Mutually exclusive outcomes separate absence of an RMSD-successful candidate, chirality exclusion, ranking failure within the effective eligible set, selected-pose PoseBusters failure, and RMSD–PoseBusters success. The “generation failure” category refers to the refined bank and does not isolate the initial sampler. If no candidate passes the mask, the effective set follows the actual unfiltered fallback. (B) Cumulative selected-pose RMSD regret relative to the minimum RMSD in that same effective set. Reference RMSD is used only for retrospective analysis, and PoseBusters validity is known only for selected poses.

S7.15 Generation-input sensitivity and guidance controls

The sensitivity protocol in Online Methods crosses five crop radii with three center-jitter levels at prior scale 2 Å, and three prior scales with the same jitter levels at a 10 Å crop. Three grid cells are shared. Four candidate-budget/guidance cells include the reused primary baseline, giving $5 \times 3 + 3 \times 3 - 3 + (4 - 1) = 24$ unique conditions in total. The same three inference seeds are used for each complex. This analysis refines and rescores saved candidate banks from those prespecified generation conditions; it does not generate additional candidates. Raw and refined banks are scored with the actual source prior scale, which also enters the docking features recomputed for confidence. Neither refinement nor confidence scoring receives the perturbed center or generation crop: both retain the original center and a 10 Å crop. These experiments therefore do not quantify end-to-end performance under a wrongly supplied pocket.

Paired conditions use the same seeds, fragment inventories and prior-draw construction. The jitter stream is independent of the pose prior; within each complex and repeat, jitter levels scale the same perturbation vector. Translation draws were identical; backend quaternion roundoff produced initial-rotation differences of at most 2.81×10^{-7} radians, without bitwise prior identity across backends. The 40- and 100-pose budgets are generated separately and are not nested banks.

Guidance controls add the capped energy drift in Eq. (S66). Guided raw-bank mean RMSD–PoseBusters success was higher in both cohorts at each budget; refined-bank differences were smaller and had mixed signs (Fig. S19). The primary protocol remains unguided.

Refinement preserves candidate identity, enabling raw–refined coordinate comparisons.

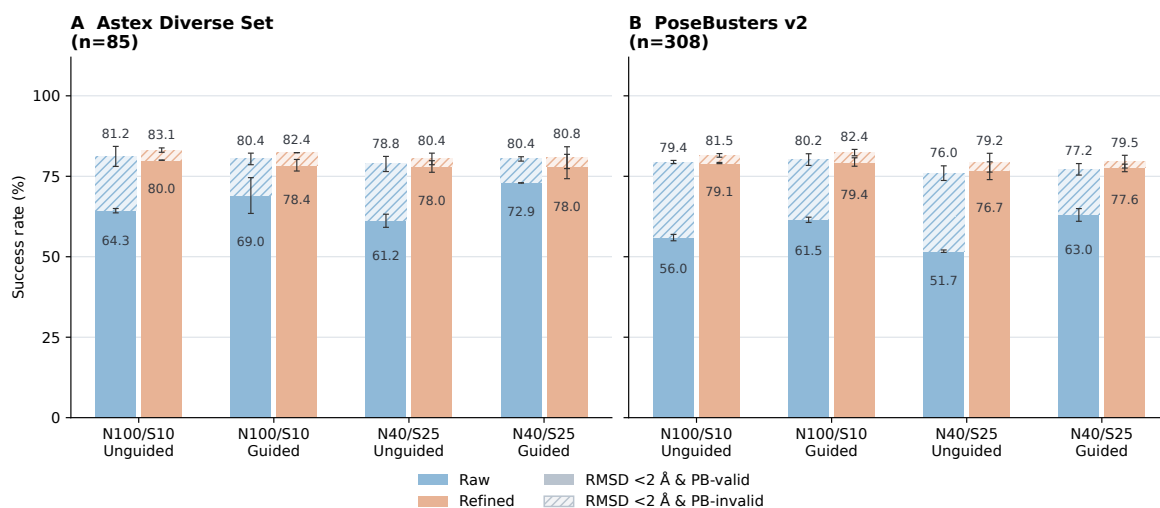


Figure S19: **Guidance and candidate allocation at a fixed learned pose-step count.** (A) Astex Diverse Set (85 complexes) and (B) PoseBusters v2 (308 complexes). Each 100-pose/ten-step or 40-pose/25-step condition uses 1,000 learned pose-steps, with unguided generation or normalized energy guidance of strength $\eta = 2$. Raw and refined banks use the input-chirality-filtered minimum-predicted-RMSD selector, with an unfiltered fallback. Solid segments show RMSD and PoseBusters success; hatched extensions show RMSD-successful but PB-invalid outcomes. Bars and whiskers show the mean and sample SD over three inference seeds. Guided and unguided initial draws are paired within each budget; the two budgets are separately executed. Equal pose-step counts are not equal-runtime comparisons.

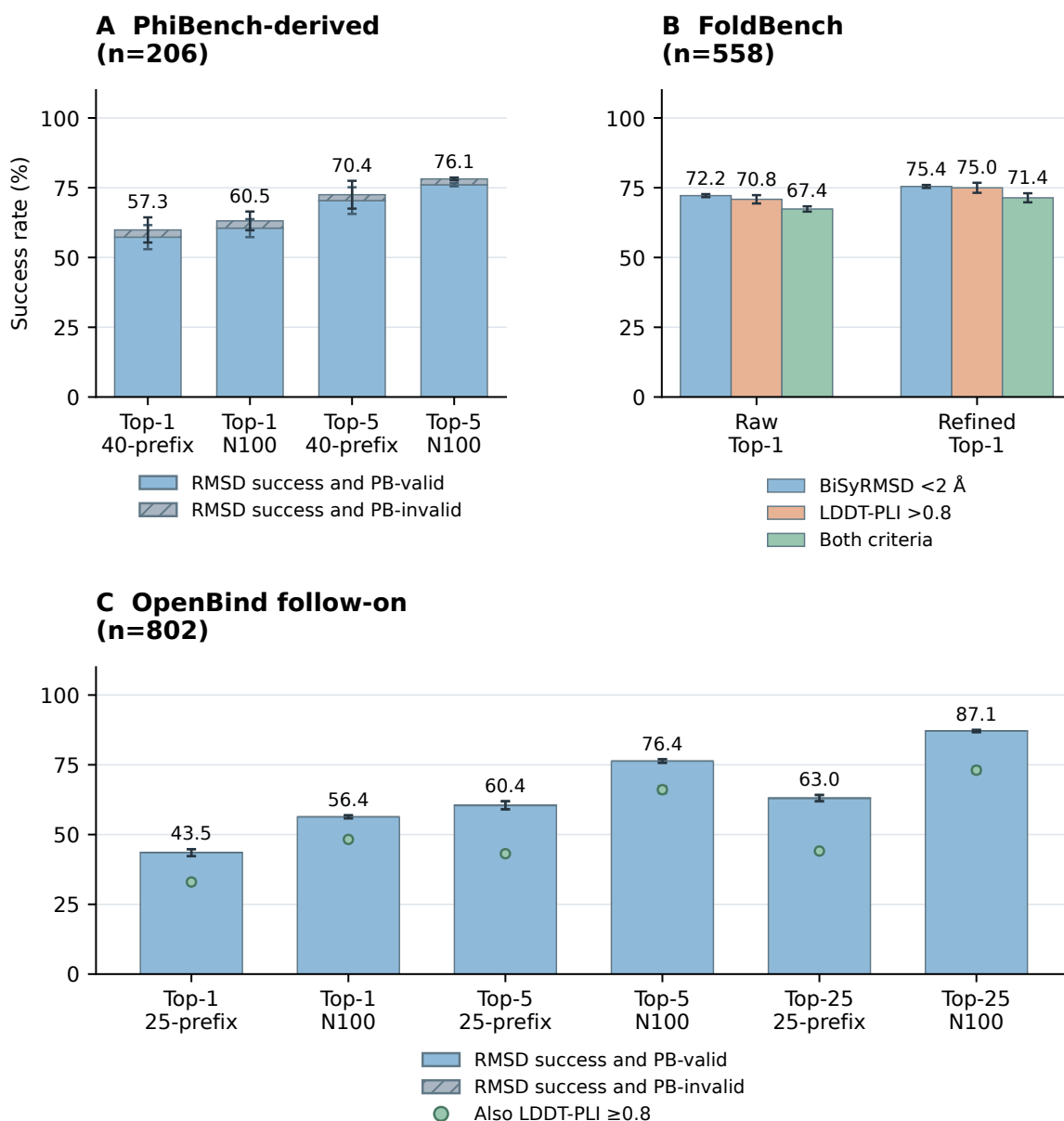


Figure S20: **Benchmark-specific evaluation of saved candidates.** Supplementary endpoint analyses of saved EFF-Dock candidates. (A) Top-1 and Top-5 fixed-receptor symmetry-corrected RMSD success ($< 2 \text{ \AA}$), with and without all 27 non-RMSD PoseBusters checks, in the locally selected PhiBench-derived cohort. This is not asserted to reproduce the native PhysDock PAL-RMSD, cohort or 18-check conjunction. (B) Raw and refined Top-1 BiSyRMSD ($< 2 \text{ \AA}$), LDDT-PLI (> 0.8), and their same-pose conjunction on all 558 FoldBench interfaces using OpenStructure 2.8.0. The supplied holo receptor makes this a redocking evaluation rather than the source cofolding task. (C) Top-1/5/25 on the official 802 OpenBind follow-on IDs, using OpenStructure 2.11.1 BiSyRMSD $\leq 2 \text{ \AA}$, PoseBusters 0.6.5, and the additional same-pose LDDT-PLI ≥ 0.8 criterion. A and C use refined banks and the frozen chirality-filtered confidence selector, including its no-eligible-candidate fallback. Hatching denotes RMSD-successful but PB-invalid cases. 40-prefix and 25-prefix restrict the original saved N100 banks before selection; they do not represent fresh smaller-budget generation or equal compute. Numbers in A and C give RMSD–PoseBusters success; numbers in B give the corresponding criterion. Bars and error lines show means and sample standard deviations across three fixed-weight inference repeats. Rates retain the entire cohort denominator; endpoint-specific determinate denominators and unresolved counts are provided in the source data. Native preparation differences remain.

References

- [1] Greg Landrum et al. RDKit: Open-source cheminformatics software. <https://www.rdkit.org>, 2024.
- [2] Shuzhe Wang, Jagna Witek, Gregory A. Landrum, and Sereina Riniker. Improving conformer generation for small rings and macrocycles based on distance geometry and experimental torsional-angle preferences. *Journal of Chemical Information and Modeling*, 60(4):2044–2058, 2020.
- [3] Thomas A. Halgren. Merck molecular force field. i. basis, form, scope, parameterization, and performance of MMFF94. *Journal of Computational Chemistry*, 17(5–6):490–519, 1996.
- [4] Wolfgang Kabsch. A solution for the best rotation to relate two sets of vectors. *Acta Crystallographica Section A*, 32(5):922–923, 1976.
- [5] Janani Durairaj, Yusuf Adeshina, Zhonglin Cao, Xuejin Zhang, Vladas Oleinikovas, et al. PLINDER: The protein–ligand interactions dataset and evaluation resource. *bioRxiv*, 2024.
- [6] Ken Shoemake. Uniform random rotations. In David Kirk, editor, *Graphics Gems III*, pages 124–132. Academic Press, 1992.
- [7] Yi-Lun Liao and Tess Smidt. Equiformer: Equivariant graph attention transformer for 3d atomistic graphs. In *International Conference on Learning Representations*, 2023.
- [8] NVIDIA. cuEquivariance, version 0.9.0. Software, 2026.
- [9] Mario Geiger and Tess Smidt. e3nn: Euclidean neural networks, 2022.
- [10] Ruiqi Feng, Chenglei Yu, Wenhao Deng, Peiyan Hu, and Tailin Wu. On the guidance of flow matching. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 16993–17029. PMLR, 2025.
- [11] Anthony K. Rappé, Carla J. Casewit, K. S. Colwell, William A. Goddard, III, and W. M. Skiff. UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations. *Journal of the American Chemical Society*, 114(25):10024–10035, 1992.
- [12] A. Bondi. van der waals volumes and radii. *The Journal of Physical Chemistry*, 68(3):441–451, 1964.
- [13] Sebastian Salentin, Sven Schreiber, V. Joachim Haupt, Melissa F. Adasme, and Michael Schroeder. PLIP: Fully automated protein–ligand interaction profiler. *Nucleic Acids Research*, 43(W1):W443–W447, 2015.
- [14] Kelly K. Lee, Carolyn A. Fitch, and Bertrand García-Moreno E. Distance dependence and salt sensitivity of pairwise, coulombic interactions in a protein. *Protein Science*, 11(5):1004–1016, 2002.
- [15] Georgia B. McGaughey, Marc Gagné, and Anthony K. Rappé. π -stacking interactions: Alive and well in proteins. *Journal of Biological Chemistry*, 273(25):15458–15463, 1998.
- [16] Justin P. Gallivan and Dennis A. Dougherty. Cation– π interactions in structural biology. *Proceedings of the National Academy of Sciences of the United States of America*, 96(17):9459–9464, 1999.
- [17] Pascal Auffinger, Franklin A. Hays, Eric Westhof, and P. Shing Ho. Halogen bonds in biological molecules. *Proceedings of the National Academy of Sciences of the United States of America*, 101(48):16789–16794, 2004.
- [18] Gautam R. Desiraju, P. Shing Ho, Lars Kloo, Anthony C. Legon, Roberto Marquardt, Pierangelo Metrangolo, Peter Politzer, Giuseppe Resnati, and Kari Rissanen. Definition of the halogen bond (IUPAC recommendations 2013). *Pure and Applied Chemistry*, 85(8):1711–1713, 2013.

- [19] Ivan Dokmanić, Mile Šikić, and Sanja Tomić. Metals in proteins: Correlation between the metal-ion type, coordination number and the amino-acid residues involved in the coordination. *Acta Crystallographica Section D: Biological Crystallography*, 64(3):257–263, 2008.
- [20] Milana Bazayeva, Claudia Andreini, and Antonio Rosato. A database overview of metal-coordination distances in metalloproteins. *Acta Crystallographica Section D: Structural Biology*, 80(5):362–376, 2024.
- [21] Geng-Yu Lin, Yu-Cheng Su, Yen Lin Huang, and Kun-Yi Hsin. MESPEUS: A database of metal coordination groups in proteins. *Nucleic Acids Research*, 52(D1):D483–D493, 2024.
- [22] Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein. Muon: An optimizer for hidden layers in neural networks. Online technical report, 2024.
- [23] Martin Buttenschoen, Garrett M. Morris, and Charlotte M. Deane. PoseBusters: AI-based docking methods fail to generate physically valid poses or generalise to novel sequences. *Chemical Science*, 15:3130–3139, 2024.
- [24] Michael J. Hartshorn, Marcel L. Verdonk, Gianni Chessari, Suzanne C. Brewerton, Wijnand T. M. Mooij, Paul N. Mortenson, and Christopher W. Murray. Diverse, high-quality test set for the validation of protein–ligand docking performance. *Journal of Medicinal Chemistry*, 50(4):726–741, 2007.
- [25] Kexin Zhang, Yuanyuan Ma, Jiale Yu, Huiting Luo, Jinyu Lin, Yifan Qin, Xiangcheng Li, Qian Jiang, Fang Bai, Jiayi Dou, et al. PhysDock: A physics-guided all-atom diffusion model for protein–ligand complex prediction. *bioRxiv*, 2025.
- [26] Sheng Xu, Qiantai Feng, Lifeng Qiao, Hao Wu, Tao Shen, Yu Cheng, Shuangjia Zheng, and Siqi Sun. Benchmarking all-atom biomolecular structure prediction with FoldBench. *Nature Communications*, 17:442, 2026.
- [27] OpenBind Consortium. OpenBind structure–affinity data release: Enterovirus A71 (EV-A71) / coxsackievirus A16 (CVA16) 2a protease, 2026.
- [28] Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J. Ballard, John M. Jumper, et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630:493–500, 2024.
- [29] John D. Westbrook, Chenghua Shao, Zukang Feng, Marina Zhuravleva, Sameer Velankar, and Jasmine Young. The chemical component dictionary: Complete descriptions of constituent molecules in experimentally determined 3d macromolecules in the protein data bank. *Bioinformatics*, 31(8):1274–1278, 2015.
- [30] Alvaro Prat, Leo Zhang, Charlotte M. Deane, Yee Whye Teh, and Garrett M. Morris. SigmaDock: Untwisting molecular docking with fragment-based SE(3) diffusion. In *The Fourteenth International Conference on Learning Representations*, 2026.
- [31] Michael Plainer, Marcella Toth, Simon Dobers, Hannes Stärk, Gabriele Corso, Céline Marquet, and Regina Barzilay. DiffDock-Pocket: Diffusion for pocket-level docking with sidechain flexibility. In *Machine Learning in Structural Biology Workshop, NeurIPS*, 2023.
- [32] J. Henry Broster, Bojana Popovic, Diana Kondinskaia, Charlotte M. Deane, and Fergus Imrie. Teaching diffusion models physics: Reinforcement learning for physically valid diffusion-based docking. *Chemical Science*, 2026.
- [33] Jintao Zhu, Zhonghui Gu, Jianfeng Pei, and Luhua Lai. DiffBindFR: An SE(3) equivariant network for flexible protein–ligand docking. *Chemical Science*, 15(21):7926–7942, 2024.
- [34] Duanhua Cao, Mingan Chen, Runze Zhang, Zhaokun Wang, Manlin Huang, Jie Yu, Xinyu Jiang, Zhehuan Fan, Wei Zhang, Hao Zhou, et al. SurfDock is a surface-informed diffusion generative model for reliable and accurate protein–ligand complex prediction. *Nature Methods*, 22:310–322, 2025.

- [35] Oleg Trott and Arthur J. Olson. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31(2):455–461, 2010.
- [36] Jerome Eberhardt, Diogo Santos-Martins, Andreas F. Tillack, and Stefano Forli. AutoDock Vina 1.2.0: New docking methods, expanded force field, and python bindings. *Journal of Chemical Information and Modeling*, 61(8):3891–3898, 2021.
- [37] Andrew T. McNutt, Yanjing Li, Rocco Meli, Rishal Aggarwal, and David Ryan Koes. GNINA 1.3: The next increment in molecular docking with deep learning. *Journal of Cheminformatics*, 17:28, 2025.
- [38] Stephen R. Heller, Alan McNaught, Igor Pletnev, Stephen Stein, and Dmitrii Tchekhovskoi. InChI, the IUPAC international chemical identifier. *Journal of Cheminformatics*, 7:23, 2015.
- [39] David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010.
- [40] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.